

CHAPTER 8

SELECTED FAMILIES OF DISTRIBUTIONS

Several most commonly used distributions were introduced in the preceding chapters. We will now present them in a systematic way, adding new information while also providing references to preceding chapters.

8.1 BERNOULLI TRIALS AND RELATED DISTRIBUTIONS

Bernoulli trials refer to independent repetitions of some experiment in which we are interested in an event A that occurs in each trial with the same probability p . We refer to event A as “success,” and the event A^c as a “failure.” The decision about which event, A or A^c , is labeled success is arbitrary and usually implies nothing about its nature in any practical applications.

The Bernoulli random variable, a building block of the theory, is just a count of the number of successes in a single trial. Thus, X is 1 or 0, depending on whether A or A^c occurred, and consequently, the distribution of X is given by

$$P\{X = 0\} = 1 - p, \quad P\{X = 1\} = p. \quad (8.1)$$

It will be convenient to let $q = 1 - p$. The expectation and variance are then

$$E(X) = p, \quad \text{Var}(X) = p(1 - p) = pq.$$

Moreover, since $0^n = 0$ and $1^n = 1$, the Bernoulli random variable is the only (nondegenerate) random variable X that satisfies the relations $X^n = X$ for all $n \geq 1$. Consequently, the moments of X are

$$m_n = E(X^n) = E(X) = p$$

for all $n \geq 1$. Then, by the fact that $\frac{d^n}{dt^n} m_X(t)|_{t=0} = p$ for all $n \geq 1$, the Taylor expansion of the moment generating function of X is

$$m_X(t) = m_0 + \sum_{n=1}^{\infty} m_n \frac{t^n}{n!} = 1 + p \sum_{n=1}^{\infty} \frac{t^n}{n!} = 1 + p(e^t - 1) = pe^t + q.$$

This result, of course, could have been obtained in a much simpler way using the distribution (8.1) of X .

Binomial Distribution

The binomial distribution plays the central role in probability theory and statistics, as a model of the total number of successes in n Bernoulli trials. In this chapter, we let S_n denote the binomial random variable, and we use the representation

$$S_n = X_1 + \cdots + X_n, \quad (8.2)$$

where $X_1, \dots, X_n$ are the Bernoulli random variables describing the outcomes of successive trials (i.e., $X_i = 1$ or 0 , depending on whether the i th trial results in success or in failure). We have encountered binomial random variables in the preceding chapters (e.g., in Examples 5.7, 6.14, 7.3, and 7.33).

Its probability distribution is

$$P\{S_n = k\} = \binom{n}{k} p^k (1-p)^{n-k} = b(k; n, p) \quad (8.3)$$

and from (7.32),

$$E(S_n) = np, \quad \text{Var}(S_n) = np(1-p).$$

The symbol $\text{BIN}(n, p)$ denotes binomial distribution with parameters n and p , so we can say that S_n has distribution $\text{BIN}(n, p)$ or simply $S_n \sim \text{BIN}(n, p)$. The Bernoulli random variable has the distribution $\text{BIN}(1, p)$.

To find the most likely number of successes (the mode), we proceed as follows: We have

$$\frac{b(k; n, p)}{b(k-1; n, p)} = \frac{\binom{n}{k} p^k (1-p)^{n-k}}{\binom{n}{k-1} p^{k-1} (1-p)^{n-k+1}} = \frac{n-k+1}{k} \times \frac{p}{q}.$$

Since $\frac{n-k+1}{k} \times \frac{p}{q} > 1$ if and only if $k < (n+1)p$, the probabilities $b(k; n, p)$ initially increase and then decrease. If $(n+1)p$ is not an integer, the unique maximum of probabilities $b(k; n, p)$ is attained at $k^* = [(n+1)p]$ (an integer part of $(n+1)p$), while if $(n+1)p$ is an integer, the maximum is attained at two values, $(n+1)p$ and $(n+1)p - 1$.

The moment generating function of binomial distribution was obtained in Example 7.22:

$$m_{S_n}(t) = E(e^{tS_n}) = \sum_{k=0}^n e^{tk} \binom{n}{k} p^k q^{n-k} = (pe^t + q)^n.$$

The same result can be obtained using Theorem 7.5.5 and formula (8.2) by observing that $m_{S_n}(t) = [m_X(t)]^n$, where $m_X(t)$ is the moment generating function of the Bernoulli random variable.

Calculating the numerical values of binomial probabilities is simple, in principle, but can be cumbersome, especially when—as is often the case—we need to know probabilities of the

form $P\{a \leq S_n \leq b\}$, which require calculating the individual probabilities $P\{S_n = k\}$ for all k between a and b .

While one could use statistical tables to obtain some binomial probabilities (especially for a small or moderate n), the most common today is using statistical software.

■ EXAMPLE 8.1

Assume that it is known (from past experience, research surveys, etc.) that 40% of buyers of FATCOW butter buy unsalted butter, and the remaining 60% buy salted butter. The store expects to sell no more than 20 packages of FATCOW butter per day, so they put on the shelf 8 packs of unsalted, and 12 packs of salted FATCOW butter. It happened that only 15 persons bought FATCOW butter on a given day, each person buying one package. What is the probability that all buyers found the kind of butter they wanted?

SOLUTION. We assume that the choices made by different people are independent (which may be a reasonable assumption, but we advise the reader to think of a situation where assumption of independence is not valid). If we let S_{15} denote the number of persons (out of 15 buyers) who bought unsalted FATCOW butter, then S_{15} has binomial distribution $\text{BIN}(15, 0.4)$. If all customers are to find the kind of butter they want, we must have $S_{15} \leq 8$ and also $15 - S_{15} \leq 12$, that is $S_{15} \geq 3$. Thus, we need $P\{3 \leq S_{15} \leq 8\}$. Rather than computing the sum $\sum_{k=3}^8 \binom{15}{k} (0.4)^k (0.6)^{15-k}$, it is more convenient to get the required probability as $P\{S_{15} \leq 8\} - P\{S_{15} \leq 2\}$ for $n = 15$ and $p = 0.4$.

■ EXAMPLE 8.2 A Warning

The probability that a Montezuma rose will blossom during the first year after planting is 80%. Mrs. Smith bought 20 bushes of Montezuma roses and planted them in her garden. What is the probability that less than 75% of her roses will blossom the first year?

SOLUTION. We regard the number S_{20} of rose bushes that will blossom in the first year as a random variable with distribution $\text{BIN}(20, 0.8)$. To obtain the probability $P\{S_{20} < 15\} = P\{S_{20} \leq 14\}$, we introduce $S'_{20} = 20 - S_{20}$ —the number of roses that do not blossom. S'_{20} has binomial distribution $\text{BIN}(20, 0.2)$, and $P\{S_{20} < 15\} = P\{20 - S'_{20} < 15\} = P\{S'_{20} > 5\} = 1 - P\{S'_{20} \leq 5\} = 1 - 0.630 = 0.370$.

The truth is, however, that the *real* answer to this problem simply cannot be computed. Indeed, the probability 0.8 of blossoming in the first year after planting presumably represents some kind of overall success average, obtained from data for various years, soil conditions, gardening techniques, and so on. The roses of Mrs. Smith are likely to be subjected to the specific conditions, such as same type of care. Consequently, the probability of success for Mrs. Smith garden need not be 0.8, and—more important—blossoming of her roses is unlikely to be independent of one another.

A more realistic assumption here might be that probability of blossoming p is random. Keeping the assumption of independence for every p , this will make the blossoming of Mrs. Smith's roses exchangeable (but not independent) events (see Chapter 4). Thus, the solution above is obtained only at the cost of accepting the assumption of independence and $p = 0.8$.

This example is placed here to make the reader aware that modeling real situations requires accepting some assumptions. Justification of these assumptions can sometimes be a complicated issue.

A difficulty with tabulating binomial distribution lies in the fact that it is necessary to have a separate table for each pair n and p , for $0 < p \leq 0.5$. Moreover, the tables become increasingly cumbersome when n increases. As we will show in subsequent sections, the situation can be remedied substantially for large n . As a preparation for the approximation introduced later in this chapter, let us consider an example.

■ EXAMPLE 8.3

Assume that about one birth in 80 is a twin birth. What is the probability that there will be no twin births among the next 200 births in the maternity ward of a given hospital?

SOLUTION. Clearly, we have here a binomial situation. If S_{200} denotes the number of twin births among the next 200 births, we need $P\{S_{200} = 0\}$, where $S_{200} \sim \text{BIN}(200, 1/80)$. Thus, remembering that $\lim_{n \rightarrow \infty} (1 - \frac{c}{n})^n = e^{-c}$, we write:

$$\begin{aligned} P\{S_{200} = 0\} &= \binom{200}{0} \left(\frac{1}{80}\right)^0 \left(\frac{79}{80}\right)^{200} = \left(1 - \frac{1}{80}\right)^{200} \\ &= \left(1 - \frac{200/80}{200}\right)^{200} \approx e^{-200/80} = 0.0821. \end{aligned} \quad (8.4)$$

The approximation works well if the probability of success p is small and number of trials n is large. Specifically, the error, equal to $|P\{S_n = 0\} - e^{-np}|$, depends on the value of the product np . As we will see, (8.4) is an example of the Poisson approximation to the binomial distribution, which we will discuss later.

Geometric Distribution

Another random variable related to Bernoulli trials is the number X of failures preceding the first success. We earlier encountered this variable in Examples 5.8 and 7.4. Since

$$P\{X = k\} = q^k p, \quad k = 0, 1, 2, \dots, \quad (8.5)$$

and probabilities

$$P\{X \geq k\} = q^k p + q^{k+1} p + \dots = p \frac{q^k}{1 - q} = q^k \quad (8.6)$$

are obtained by summing the geometric series, the distribution (8.5) is called *geometric*. Based on formula (8.6), one obtains the cdf formula, since

$$F_X(k) = P(X \leq k) = 1 - q^{k+1}.$$

A distribution closely associated with (8.5) is that of the number Y of trials up to and including the first success, which we write as

$$P\{Y = n\} = P\{X = n - 1\} = q^{n-1} p, \quad n = 1, 2, \dots \quad (8.7)$$

Actually, both distributions, (8.5) and (8.7), are called geometric in the literature. Since $Y = X + 1$, these distributions are closely related, and most results are valid under either definition. We will use both definitions, specifying (in cases where it makes a difference) whether (8.5) or (8.7) is used. We introduce both definitions deliberately, not in order to confuse our readers but to make them flexible in using terminology that is not well established. We will also use the same symbol for both distributions, (8.5) and (8.7). In other words, we will use the notation $\text{GEO}(p)$ to denote the geometric distribution with probability of success being p .

The expected value of the geometric distribution was found in Example 7.5. For the distribution (8.7), we have

$$E(Y) = \frac{1}{p}, \quad (8.8)$$

so for the distribution (8.5), we obtain

$$E(X) = E(Y - 1) = \frac{1}{p} - 1 = \frac{q}{p}. \quad (8.9)$$

The variance of the geometric distribution is the same for both X and Y (see Theorem 7.6.1),

$$\text{Var}(X) = \text{Var}(Y) = \frac{q}{p^2}. \quad (8.10)$$

It can be derived using Theorem 7.5.3 and the mgf of the geometric distribution

$$m_X(t) = \frac{p}{1 - qe^t} \quad \text{and} \quad m_Y(t) = m_{X+1}(t) = \frac{pe^t}{1 - qe^t}$$

for $qe^t < 1$; hence, for $t < -\log q$.

Let us observe that for all $m = 0, 1, 2, \dots$ and $n = 0, 1, 2, \dots$

$$P\{X \geq m + n | X \geq m\} = P\{X \geq n\}. \quad (8.11)$$

Indeed, since $X \geq m + n$ implies that $X \geq m$, we can write

$$P\{X \geq m + n | X \geq m\} = \frac{P\{X \geq m + n\}}{P\{X \geq m\}} = \frac{q^{m+n}}{q^m} = q^n = P\{X \geq n\}.$$

Formula (8.11) is said to express the *memoryless property* of geometric distribution: if waiting time for the first success is at least m , then the probability that it will be at least $m + n$ is the same as the probability that the waiting time for the first success will be at least n .

Formula (8.11) is valid for random variable X . For the random variable Y (number of trials up to and including the first success), we have a formula with the strict inequalities

$$P\{Y > m + n | Y > m\} = P\{Y > n\}. \quad (8.12)$$

Formula (8.11) characterizes geometric distribution (8.5); equivalently, (8.12) characterizes distribution (8.7). We will prove

Theorem 8.1.1 *If Y is a discrete random variable that may assume values $1, 2, \dots$ and satisfies (8.12), then Y has the distribution (8.7) for some p and $q = 1 - p$.*

Proof: Let $\pi_k = P\{Y = k\}$ and $\eta_k = \pi_{k+1} + \pi_{k+2} + \dots = P\{Y > k\}$. Then (8.12) means that we must have $\eta_{m+n}/\eta_m = \eta_n$ for all $m, n = 1, 2, \dots$. In particular, $\eta_2 = \eta_1^2$, and by induction, $\eta_k = \eta_1^k$. Consequently, $\pi_k = \eta_k - \eta_{k+1} = \eta_1^k - \eta_1^{k+1} = \eta_1^k(1 - \eta_1)$ for all k , which means that Y has geometric distribution with $q = \eta_1 = P\{Y > 1\}$. $\square$

It is sufficient to require that (8.12) holds for $m = 1$ and all n or for $n = 1$ and for all m . Also the Theorem 8.1.1 holds if m is replaced by a random variable (see Srivastava, 1981).

Theorem 8.1.2 *Let U be a random variable which assumes only positive integer values. If*

$$P\{Y > U + n | Y > U\} = P\{Y > n\}, \quad (8.13)$$

then Y has geometric distribution (8.7) for some p and $q = 1 - p$.

■ EXAMPLE 8.4

In Section 4.7 we discussed Markov chains, these being processes that describe the evolution of a system that may at any time be in one of a specified set of states and may change its state at times $t = 1, 2, \dots$. The main assumption is that if at some time t the system is in state i , then the probability that it will pass to state j at time $t + 1$ is p_{ij} , regardless of the history of transitions prior to time t . This means that given the present (state at time t), the future and the past are independent. Suppose now that at some time t the system enters state i , and let T_i be the duration of the current visit to state i . More precisely, we define $T_i = k$ if the state at times $t, t + 1, t + 2, \dots, t + k$ is i , but the state at time $t + k + 1$ is different from i . We have then

$$P\{T_i = k\} = p_{ii}^k(1 - p_{ii}), \quad k = 1, 2, \dots,$$

and we see that the duration of a visit in state i has geometric distribution with $q = p_{ii}$ and $p = 1 - p_{ii}$.

The memoryless property of geometric distribution is in fact the Markov property. Indeed, suppose that the system stayed in state i for m units of time prior to time t . Then the probability of staying there for at least n additional units is the same as the probability of staying in state i for at least $m + n$ units of time. Given the present state, the future (in particular, the duration of remaining stay in the present state) is independent of the past (in particular, of how long the system has already stayed in its present state).

Theorem 8.1.2 asserts a seemingly identical property, but a moment of reflection shows that the condition is now much stronger: the “present” is not some fixed time, but it is a time that may be random. In particular, this randomness may be affected by the process itself.

To give an example, let $X = T_i$, the time of remaining in the state i just entered, and let U be the longest visit in state i recorded thus far (so that U depends on the history of the process up to the present time). Theorem 8.1.2 asserts, in particular, the following: given that the duration of the present state in state i will break the record ($T_i > U$), the probability of breaking the record by at least 3 units (i.e., $T_i > U + 3$) is the same as the probability of a visit lasting longer than 3 units ($T_i > 3$). This property stands in contrast with what one observes in sport records: the consecutive improvements of the world record in any discipline tend to be smaller.

The property asserted in Theorem 8.1.2 is generally called *strong* Markov property. It means that given the present, the future is independent of the past, even if the “present” is selected at random, as long as this randomness depends only on the past, and not on the future.

Theorems 8.1.1 and 8.1.2 are examples of *characterizations* of probability distributions. This kind of theorem singles out a certain property of a type of distribution and shows that this property is valid only for distributions of this type (in the case above, the memoryless property for the geometric distribution).

The characterization theorems are of a great practical value. As we will see in the later chapters, if one is interested in more than merely summarizing and presenting the statistical data, it is useful to regard the data (results of observations, experiments, surveys, etc.) as values of some random variables. Roughly speaking, the more we know about the distributions of these random variables, the better is our understanding of the phenomenon studied, as well as the better are our possibilities of prediction and control.

To fix the idea, imagine that a statistician’s data can be regarded as independent observations of some random variable X . It happens quite often that identification of the distribution of X is accomplished in two steps. The first step consists of identifying a class of

distributions that contains the distribution of X . The second step is the identification of the particular distribution in this class.

In many cases, however, determining the class of distributions that contains distribution of X is not so simple. In such situations, one can sometimes use a characterization theorem: if one knows that the distribution has some property that turns out to be characteristic for a given class, then the distribution must be in that class (e.g., if one knows that the distribution of X is memoryless, then it must be geometric).

The following example shows how the geometric distribution can be used for modeling.

■ EXAMPLE 8.5 Family Planning

Assume that the sexes of consecutive children born to the same parents are independent, with the probability of a child being a girl equal to π . We will disregard twin births and consider family sizes under various plans.

Plan 1. The couple decides to stop having children as soon as their first girl is born. Let ξ_1 be the number of children according to this plan. Then ξ_1 is a geometric random variable with $P\{\xi_1 = n\} = (1 - \pi)^{n-1}\pi$ so that $E(\xi_1) = 1/\pi$. If the chances of a child being a boy are the same as those of being a girl, we have $\pi = 0.5$ and $E(\xi_1) = 2$.

Plan 2. The couple decides to have children until they have a boy and a girl and then stop. Let ξ_2 be the number of children under this plan. To determine $P(\xi_2 = n)$, let G and B denote the event “first child is a girl (boy).” Then, by the total probability formula (4.9), for $n \geq 2$ we have

$$\begin{aligned} P(\xi_2 = n) &= \pi P(\xi_2 = n|G) + (1 - \pi)P(\xi_2 = n|B) \\ &= \pi P\{Y^B = n - 1\} + (1 - \pi)P\{Y^G = n - 1\}. \end{aligned}$$

Here Y^B and Y^G are the numbers of children (excluding the first) that a family will have until the first boy (girl) is born. Clearly, Y^B and Y^G have the distribution given by (8.6), with probability of success being $1 - \pi$ for Y^B and π for Y^G . Thus, for $n = 2, 3, \dots$,

$$\begin{aligned} P\{\xi_2 = n\} &= \pi[\pi^{n-2}(1 - \pi)] + (1 - \pi)[(1 - \pi)^{n-2}\pi] \\ &= \pi^{n-1}(1 - \pi) + (1 - \pi)^{n-1}\pi. \end{aligned}$$

To find the expected number of children under plan 2, we write

$$E(\xi_2) = \pi \sum_{n=2}^{\infty} nP\{Y^B = n - 1\} + (1 - \pi) \sum_{n=2}^{\infty} nP\{Y^G = n - 1\},$$

and, putting $k = n - 1$, obtain

$$\begin{aligned} \sum_{n=2}^{\infty} nP\{Y^B = n - 1\} &= \sum_{n=2}^{\infty} [(n - 1) + 1]P\{Y^B = n - 1\} \\ &= \sum_{k=1}^{\infty} kP\{Y^B = k\} + \sum_{k=1}^{\infty} P\{Y^B = k\} \\ &= \frac{1}{1 - \pi} + 1. \end{aligned}$$

By symmetry, $\sum_{n=2}^{\infty} nP\{Y^G = n-1\} = 1/\pi + 1$, and finally

$$\begin{aligned} E(\xi_2) &= \pi \left(\frac{1}{1-\pi} + 1 \right) + (1-\pi) \left(\frac{1}{\pi} + 1 \right) = \frac{\pi}{1-\pi} + \frac{1-\pi}{\pi} + 1 \\ &= \frac{1}{\pi(1-\pi)} - 1. \end{aligned}$$

Again, if $\pi = 1/2$, then $E(\xi_2) = 3$.

Negative Binomial Distribution

The geometric distribution allows us to make an immediate and natural extension. Rather than to consider the number of Bernoulli trials up to the first success, we can consider the number Y of Bernoulli trials up to and including the r th success. In analogy with the geometric distribution, we will also consider the random variable X defined as the number of failures preceding the r th success.

We will start by deriving the probability distribution of the random variables X and Y . Clearly, the possible values of Y are integers $r, r+1, r+2, \dots$. The event $\{Y = n\}$ occurs if:

1. The n th trial results in success.
2. The first $n-1$ trials give exactly $r-1$ successes (and $n-r$ failures).

Indeed, the conjunction of 1 and 2 ensures that the r th success occurs at trial n . The events 1 and 2 are independent (since their occurrence is determined by disjoint sets of trials), and their probabilities are p and $\binom{n-1}{r-1} p^{r-1} q^{n-r}$, respectively. Consequently,

$$P\{Y = n\} = \binom{n-1}{r-1} p^r q^{n-r} \quad (8.14)$$

for $n = r, r+1, r+2, \dots$. The number X of failures preceding the r th success is such that $X + r = Y$. So for $k = 0, 1, 2, \dots$, we have

$$P\{X = k\} = P\{Y = k+r\} = \binom{k+r-1}{r-1} p^r q^k. \quad (8.15)$$

Both random variables X and Y are referred to as having a *negative binomial* distributions. In either case, the distribution depends on two parameters, r and p . We will use the symbol $\text{NBIN}(r, p)$ to denote the negative binomial distribution, whether (8.14) or (8.15).

Let us verify that (8.15) and (8.14) are probability distributions. We must show that

$$\sum_{k=0}^{\infty} \binom{k+r-1}{r-1} p^r q^k = 1. \quad (8.16)$$

Let us observe first that by formulas (3.14) and (3.9),

$$\begin{aligned} \binom{k+r-1}{r-1} &= \binom{r-1+k}{k} \\ &= \frac{(r-1+k)(r-1+k-1)\cdots(r-1+k-k+1)}{k!} \\ &= \frac{r(r+1)\cdots(r+k-1)}{k!} \\ &= \frac{(-1)^k (-r)(-r-1)\cdots(-r-k+1)}{k!} = (-1)^k \binom{-r}{k}. \end{aligned}$$

Thus, using Newton's formula (3.16), we have

$$\sum_{k=0}^{\infty} \binom{k+r-1}{r-1} p^r q^k = p^r \sum_{k=0}^{\infty} (-1)^k \binom{-r}{k} q^k = p^r (1-q)^{-r} = 1.$$

Obviously, the proof that

$$\sum_{n=r}^{\infty} \binom{n-1}{r-1} p^r q^{n-r} = 1$$

is similar, and we can omit the details.

To determine the mean and variance of a random variable with negative binomial distribution, one can proceed in several ways, of which we will demonstrate two. The first uses direct calculations, while the second uses a representation as a sum of simpler random variables.

Observe that since $Y = X + r$, we have $E(Y) = E(X) + r$ and $\text{Var}(Y) = \text{Var}(X)$ (the latter property being a consequence of Theorem 7.6.1). Thus, it suffices to study one of the random variables X and Y .

We begin by finding the mgf of X . Proceeding as in the proof of (8.16), we have

$$\begin{aligned} m_X(t) &= Ee^{Xt} = \sum_{k=0}^{\infty} e^{kt} \binom{k+r-1}{r-1} p^r q^k = p^r \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} (qe^t)^k \\ &= p^r \sum_{k=0}^{\infty} \binom{-r}{k} (-qe^t)^k = \frac{p^r}{(1-qe^t)^r}, \end{aligned}$$

provided that $|-qe^t| < 1$, or equivalently $t < \log(1/q)$.

Consequently, $E(X) = m'_X(0) = rq/p$, and

$$E(Y) = r + E(X) = \frac{r}{p}. \quad (8.17)$$

After some algebra, we also obtain

$$\text{Var}(X) = \text{Var}(Y) = m''_X(0) - [E(X)]^2 = \frac{rq}{p^2}. \quad (8.18)$$

■ EXAMPLE 8.6

A salesman calls prospective buyers to make a sales pitch. Assume that the outcomes of consecutive calls are independent, and that on each call, he has 15% chance of making a sale. His daily goal is to make 3 sales, and he can make only 20 calls in a day. What is the probability that he achieves his goal in 18 trials? What is the probability that he does not achieve his daily goal?

SOLUTION. The “strategy” of solving a problem like this is to identify the type of distribution to be analyzed. Assuming that we identify the situation as Bernoulli trials (i.e., repeated independent trials with the same probability p of success), the problem most typically concerns either a binomial or negative binomial distribution. The crucial question here is: Is the number of trials fixed (and then the number of successes is random), or is the number of successes fixed (and the number of trials is random)? In the first case, we have the binomial distribution, and in the second case, the negative binomial distribution.

For our salesman, we want the probability that his third sale (success) comes at trial 18. Thus, the number of successes $r = 3$ is fixed (this is the salesman's goal), and it

is the number of calls that is random. We have $p = 0.15$, and we ask for $P\{X = 18\}$, where X is the number of trials up to and including the third success. Substitution to formula (8.14) gives

$$P\{X = 18\} = \binom{17}{2} (0.15)^3 (0.85)^{15} = 0.0401.$$

The second question is about the probability of the salesman not attaining his daily goal. Here the number of trials is fixed ($n = 20$), and we can treat the problem as involving the binomial distribution. Thus, if S_{20} is the number of successes in 20 trials, the salesman does not achieve his goal if $S_{20} \leq 2$. The answer is therefore

$$\begin{aligned} P\{S_{20} \leq 2\} &= P\{S_{20} = 0\} + P\{S_{20} = 1\} + P\{S_{20} = 2\} \\ &= \sum_{k=0}^2 \binom{20}{k} (0.15)^k (0.85)^{20-k} = 0.4049. \end{aligned}$$

We can also use the negative binomial distribution here. If X is the number of trials up to and including the third success, then the salesman does not attain his goal when $X > 20$. So the answer is

$$P\{X > 20\} = \sum_{n=21}^{\infty} P\{X = n\} = \sum_{n=21}^{\infty} \binom{n-1}{2} (0.15)^3 (0.85)^{n-3},$$

a quantity that is much harder to evaluate numerically than $P\{S_{20} \leq 2\}$.

The example above suggests that the probabilities in binomial and negative binomial distributions with the same probability p of success are related. The following theorem provides a useful identity:

Theorem 8.1.3 *Let $S_n \sim \text{BIN}(n, p)$ be the number of successes in n Bernoulli trials and let $Y^{(r)} \sim \text{NBIN}(r, p)$ be the number of trials up to and including the r th success. Then, for every $k = 0, 1, \dots$,*

$$P\{Y^{(r)} > k\} = P\{S_k < r\}. \quad (8.19)$$

Proof: Observe that both sides of (8.19) refer to the probability of the same event: the waiting time for the r th success exceeds k if and only if fewer than r successes occur in the first k trials. $\square$

A word of warning: Since $Y^{(r)}$ and S_k are both discrete random variables, it matters whether or not the inequalities are strict. Thus, (8.19) can be written in any of the forms, such as

$$P\{Y^{(r)} > k\} = P\{S_k \leq r - 1\}, \quad P\{Y^{(r)} < k + 1\} = P\{S_k \geq r\}.$$

An inspection of the proof of the formula (8.16) shows that the fact that r was an integer was never used. Formally, a negative binomial distribution is defined for any $r > 0$ and $0 < p < 1$ (although the interpretation of the probabilities in terms of Bernoulli trials is no longer valid).

Let us also note that the mgf of the negative binomial distribution is the r th power of the mgf of the geometric distribution. For integer values of r , this fact shows that the random variable $X^{(r)}$ (respectively, $Y^{(r)}$) is a sum of r independent geometric random variables (of

the form X or Y , depending on whether we represent $X^{(r)}$ or $Y^{(r)}$. This representation (e.g., in the case of $X^{(r)}$) means that

$$X^{(r)} = X_1 + \cdots + X_r,$$

where X_i is interpreted as the number of failures falling between the $(i-1)$ st and i th success (e.g., in case of $r=3$, if the consecutive trials are FFFSSFFFFS, then $X^{(3)}=7$ = number of failures preceding the third success, with $X_1=3$, $X_2=0$, $X_3=4$ being the numbers of failures between the three consecutive successes).

Consequently,

$$E(X^{(r)}) = E(X_1) + \cdots + E(X_r)$$

and (because of independence)

$$\text{Var}(X^{(r)}) = \text{Var}(X_1) + \cdots + \text{Var}(X_r),$$

which in view of (8.8) and (8.10) gives formulas (8.17) and (8.18). The representation of random variables $Y^{(r)}$ is analogous and will be omitted.

PROBLEMS

8.1.1 Label statements below as true or false:

(i) Suppose that 6% of all cars in a given city are Toyotas. Then the probability that there are 4 Toyotas in a row of 12 cars parked in the municipal parking is $\binom{12}{4} (0.06)^4 (0.94)^8$. (ii) Suppose that 6% of all Europeans are French. Then the probability that in a random sample of 12 inhabitants of a major European capital there are 4 Frenchmen is $\binom{12}{4} (0.06)^4 (0.94)^8$.

8.1.2 Assume that we score $Y = 1$ for a success and $Y = -1$ for a failure. Express Y as a function of the number X of successes in a single Bernoulli trial, and find moments $E(Y^n)$, $n = 1, 2, \dots$.

8.1.3 Suppose that random variables $X_1, \dots, X_n$ are independent, each with the same Bernoulli distribution. Given that $\sum_{i=1}^n X_i = r$, find: (i) The probability that $X_1 = 1$. (ii) The covariance between X_i and X_j , $1 \leq i < j \leq n$.

8.1.4 Assume that X_1 and X_2 are independent random variables, with $\text{BIN}(n_1, p)$ and $\text{BIN}(n_2, p)$ distributions, respectively. Find a correlation coefficient between X_1 and $X_1 + X_2$.

8.1.5 An experiment consists of tossing a fair coin 13 times. Such an experiment is repeated 17 times. Find the probability that in a majority of repetitions of the experiment the tails will be in minority.

8.1.6 Two balanced coins are tossed together independently. Find: (i) the distribution of the number of tosses required for both coins to show the same side; and (ii) the distribution of the number of tosses needed to get at least one head.

8.1.7 A regular die is tossed until the same side turns up twice in a row. Find: (i) the probability that it will happen in tosses 8 and 9. (ii) The probability that it will happen in tosses n and $n+1$.

8.1.8 Two players (or two teams) are negotiating the rules for determining the championship. The two possibilities are “best of five” or “best of seven.” This means that whoever wins three (respectively, four) games is the champion. Assume that games are independent and that the probability of winning a game by the first player (there

are no ties) is p . For what values of p , should the first player favor the scheme “best out of five”?

8.1.9 Show that if S_n is a binomial random variable, then for $k = 1, 2, \dots, n$,

$$P\{S_n = k\} = \frac{(n - k + 1)p}{k(1 - p)} P\{S_n = k - 1\}.$$

8.1.10 Show that for the binomial distribution we have

$$P\{S_n \leq k\} = (n - k) \binom{n}{k} \int_0^{1-p} x^{n-k-1} (1 - x)^k dx.$$

8.1.11 Six dice are tossed simultaneously until, for the first time, all of them show the same face. Find $E(U)$ and $\text{Var}(U)$, where U is the number of tosses until this happens.

8.1.12 Assume that the probability of twins being identical is β , and that the sexes of children are determined independently, with probability of a boy being b (possibly $b \neq 1/2$). Find the expected number of twin births recorded in the hospital before the first pair of: (i) boys, (ii) girls, and (iii) different genders. Note that identical twins must be of the same sex.

8.1.13 Assume that the probability that a birth is a multiple one (twins, triplets, etc.) is π . Given that a birth is a multiple one, probabilities $\alpha_2, \alpha_3, \dots$ of twins, triplets, ... satisfy the condition $\alpha_{k+1} = \gamma \alpha_k$. Find π, α_2 , and γ if it is known that the expected number of children born in 100 births is $100 + c$, and the expected number of single births observed before recording a multiple birth is M (assume that M and c given).

8.1.14 A hospital needs 20 volunteers for the control group in testing the efficiency of some treatment. The candidates are subject to psychological and medical screening, and on average, only 1 in 15 candidates is found acceptable for the experiment. The cost of screening, whether or not a candidate is found acceptable, is \$50 per person. The granting agency argues that one needs about 50×15 dollars to find one acceptable candidate, and therefore allows $20 \times 50 \times 15 = 15,000$ dollars for the cost of screening. (i) Write the formula for the probability that the allocated sum will be enough to find 20 acceptable candidates. (ii) Use the Chebyshev inequality to find the sum that gives at least a 90% chance of finding 20 acceptable candidates before the testing money runs out.

8.1.15 Assume that in a tennis match between A and B, the probability of winning a tennis set by player A is p and that the results of sets are independent. Let T be the number of sets played in a match. Find the distribution of T and $E(T)$ as a function of p , assuming that the match is played by: (i) men, (ii) women. (Note that men play “best out of five,” while women play “best out of three” sets.)

8.1.16 In the flowchart in Figure 8.1, $m > 0$ is an integer and $0 < p < 1$. The block “sample U ” means that a value of a random variable U is sampled from the $U[0, 1]$ distribution, with consecutive samplings being independent. Find the distribution of X , and then calculate its mean and variance.

8.1.17 Show that if X has a negative binomial distribution $\text{NBIN}(r, p)$, then

$$E[(r - 1)/(r + X - 1)] = p.$$

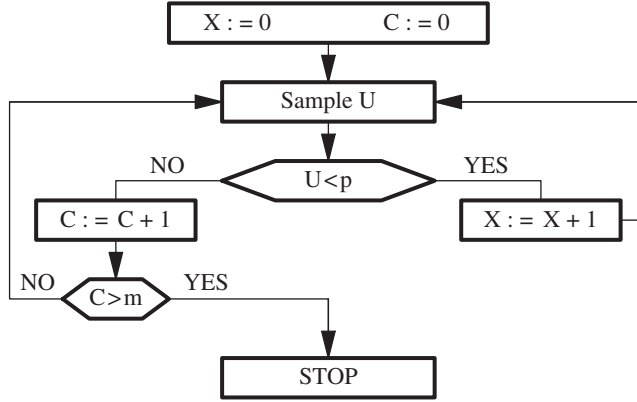


Figure 8.1 Flowchart.

8.2 HYPERGEOMETRIC DISTRIBUTION

The distributions discussed in the previous section were based on the notion of a Bernoulli trial—an event that results in a “success” or a “failure.” One of the necessary assumptions of Bernoulli trials is that the probabilities of a “success” (or equivalently “failure”) are the same in each trial. That requires either sampling from an *infinite* (practically very large) population or returning an element after it was selected (sampling with replacement). There are, however, situations where neither the population is large nor the replacing can be done. We then have sampling without replacement, with the probabilities changing after every selection and each time depending on the type of the element recovery as often used for statistical modeling (e.g., in zoology to study small animal or plant populations). We have already encountered hypergeometric distribution in Examples 3.7 and 3.8.

To proceed systematically, we assume that initially the population has N elements of which a are of one kind (“successes”) and $b = N - a$ are of another kind (“failures”). We sample n elements without replacement, and we let X denote the number of successes in the sample. Let us first determine the range of the random variable X . Clearly, $0 \leq X \leq a$, and a similar inequality for the number of failures is $0 \leq n - X \leq N - a$. This yields

$$\max(0, n - (N - a)) \leq X \leq \min(n, a). \quad (8.20)$$

If k satisfies the constraint (8.20), then

$$P\{X = k\} = \frac{\binom{a}{k} \binom{N-a}{n-k}}{\binom{N}{n}}. \quad (8.21)$$

To check that the probabilities in (8.21) add up to 1, we will use a trick. Let us consider the identity

$$(1 + x)^N = (1 + x)^a (1 + x)^{N-a}$$

and compare the coefficients of x^n on both sides. On the left-hand side the coefficient is $\binom{N}{n}$. On the right-hand side the coefficient equals the sum of all possible terms of the form $\binom{a}{k} \binom{N-a}{n-k}$, where k must satisfy the constraint (8.20). This shows that the sum of all terms (8.21) is 1, as asserted.

■ EXAMPLE 8.7

A class consists of 10 boys and 12 girls. The teacher selects 6 children at random for some task. Is it more likely that she chooses 3 boys and 3 girls or that she chooses 2 boys and 4 girls?

SOLUTION. If X is the number of boys in the selected set of children, then

$$P\{X = 3\} = \frac{\binom{10}{3} \binom{12}{3}}{\binom{22}{6}} = 0.3538, \quad P\{X = 2\} = \frac{\binom{10}{2} \binom{12}{4}}{\binom{22}{6}} = 0.2985.$$

So the first probability is higher than the second.

As expected, for large populations, it does not matter too much whether we sample with or without replacement. We shall now formulate the corresponding approximation theorem:

Theorem 8.2.1 *Let $N \rightarrow \infty$ and $a \rightarrow \infty$ in such a way that $a/N \rightarrow p$, where $0 < p < 1$. Then for every fixed n and $k = 0, 1, \dots, n$ we have*

$$\frac{P\{X = k\}}{b(k; n, p)} \rightarrow 1 \quad (8.22)$$

as $N \rightarrow \infty$. Here $P\{X = k\}$ is the probability (8.21) for hypergeometric distribution, while $b(k; n, p)$ is the binomial probability given by (8.3).

Proof: Letting J_N denote the ratio in (8.22), we write

$$\begin{aligned} J_N &= \frac{\binom{a}{k} \binom{N-a}{n-k}}{\binom{N}{n} \binom{n}{k} p^k (1-p)^{n-k}} \\ &= \frac{a!(N-a)!(N-n)!}{(a-k)!(N-a-n+k)!N!p^k(1-p)^{n-k}}, \end{aligned}$$

which cancels to

$$\frac{[a(a-1) \cdots (a-k+1)][(N-a)(N-a-1) \cdots (N-a-n+k+1)]}{N(N-1) \cdots (N-n+1)p^k(1-p)^{n-k}}.$$

Multiplying and dividing by N^n , we obtain $J_N = A_N B_N / C_N$, where

$$\begin{aligned} A_N &= \frac{a}{N} \left(\frac{a}{N} - \frac{1}{N} \right) \cdots \left(\frac{a}{N} - \frac{k-1}{N} \right), \\ B_N &= \left(1 - \frac{a}{N} \right) \left(1 - \frac{a}{N} - \frac{1}{N} \right) \cdots \left(1 - \frac{a}{N} - \frac{n-k-1}{N} \right), \\ C_N &= \left(1 - \frac{1}{N} \right) \left(1 - \frac{2}{N} \right) \cdots \left(1 - \frac{n-1}{N} \right) p^k (1-p)^{n-k}. \end{aligned}$$

Since n and k are fixed and moreover $a/N \rightarrow p$, $A_N \rightarrow p^k$, $B_N \rightarrow (1-p)^{n-k}$, and $(1-1/N) \times \cdots \times (1-(n-1)/N) \rightarrow 1$, we have $J_N \rightarrow 1$. This proves the theorem. $\square$

The practical use of Theorem 8.2.1 consists mostly of replacing probabilities

$$P\{X = k\} = \frac{\binom{a}{k} \binom{N-a}{n-k}}{\binom{N}{n}}$$

by their approximations $\binom{n}{k} \left(\frac{a}{N}\right)^k \left(1 - \frac{a}{N}\right)^{n-k}$, which are much easier to compute. The relative error of such approximation depends on all four factors, N, n, a , and k .

Let us now find the expectation and variance of the hypergeometric random variable X with distribution (8.21). We will use the method that we already used in the case of binomial and negative binomial random variables, by representing X as a sum of simpler random variables. We let

$$X = \xi_1 + \xi_2 + \cdots + \xi_n, \quad (8.23)$$

where ξ_i equals 1 or 0, depending on whether the i th draw results in success or failure.

Observe that $\xi_1, \dots, \xi_n$ are dependent random variables. This fact will affect our calculation of the variance of X . Generally, we will use the formulas

$$\begin{aligned} E(X) &= E(\xi_1) + \cdots + E(\xi_n), \\ \text{Var}(X) &= \text{Var}(\xi_1) + \cdots + \text{Var}(\xi_n) + 2 \sum_{i < j} \text{Cov}(\xi_i, \xi_j). \end{aligned}$$

Note that ξ_i 's are Bernoulli random variables (i.e., they assume only values 0 and 1). So letting $p_i = P\{\xi_i = 1\}$, we have

$$E(\xi_i) = p_i, \text{Var}(\xi_i) = p_i(1 - p_i). \quad (8.24)$$

Observe that $\xi_i \xi_j$ is also a Bernoulli random variable. So letting $p_{ij} = P\{\xi_i = 1, \xi_j = 1\}$ we have

$$\text{Cov}(\xi_i, \xi_j) = E(\xi_i \xi_j) - E(\xi_i)E(\xi_j) = P\{\xi_i \xi_j = 1\} - p_i p_j = p_{ij} - p_i p_j. \quad (8.25)$$

To determine the probabilities p_{ij} , we need the joint distribution of (ξ_i, ξ_j) . Generally, one of the ways of visualizing the distribution of the $(\xi_1, \dots, \xi_n)$ is as follows: The elements of the population (of a successes and $N - a$ failures) are ordered at random. This gives the sample space of $N!$ permutations, all of them being equally likely. The random vector $(\xi_1, \dots, \xi_n)$ is then defined at any sampled point (permutation) as the initial n elements of this permutation.

It is now clear that the joint distribution of (ξ_i, ξ_j) does not depend on (i, j) , meaning it is the same as the joint distribution of (ξ_1, ξ_2) . Indeed, the probability that $(\xi_i = x, \xi_j = y)$ depends on the number of permutations in the sample space that have specific elements x and y at places i and j . By symmetry, this number is the same as the number of permutations that have elements x and y in the first two places (or in any other designated pair of places).

Consequently, the marginal distributions of ξ_i are the same, and for all i , we must have

$$p_i = P\{\xi_i = 1\} = \frac{a}{N}.$$

Similarly

$$p_{ij} = P\{\xi_1 = 1, \xi_2 = 1\} = P\{\xi_1 = 1\} \times P\{\xi_2 = 1 | \xi_1 = 1\} = \frac{a}{N} \times \frac{a-1}{N-1}.$$

From (8.24) and (8.25) we get $E(\xi_i) = a/N$, $\text{Var}(\xi_i) = (a/N)(1 - a/N) = a(N - a)/N^2$. Therefore

$$\begin{aligned}\text{Cov}(\xi_i, \xi_j) &= E(\xi_i \xi_j) - E(\xi_i)E(\xi_j) = p_{ij} - E(\xi_i)E(\xi_j) \\ &= \frac{a}{N} \times \frac{a-1}{N-1} - \left(\frac{a}{N}\right)^2 = \frac{Na(a-1) - a^2(N-1)}{N^2(N-1)} \\ &= -\frac{a(N-a)}{N^2(N-1)} = -\frac{\text{Var}(\xi_i)}{N-1}.\end{aligned}$$

Thus $E(X) = nE(\xi_i) = n(a/N)$. For variance, observe that the number of pairs (i, j) with $i < j$ is $\binom{n}{2} = n(n-1)/2$. Consequently,

$$\begin{aligned}\text{Var}(X) &= n \times \frac{a}{N} \times \frac{N-a}{N} - 2 \times \frac{n(n-1)}{2} \times \frac{1}{N} \times \frac{a(N-a)}{N(N-1)} \\ &= n \times \frac{a}{N} \times \frac{N-a}{N} \left(1 - \frac{n-1}{N-1}\right).\end{aligned}$$

Letting $a/N = p$ denote the probability of success prior to selecting the first element, we have the following theorem:

Theorem 8.2.2 *If X has the hypergeometric distribution (8.21), then*

$$\begin{aligned}E(X) &= n \times \frac{a}{N} = np, \\ \text{Var}(X) &= n \times \frac{a}{N} \times \left(1 - \frac{a}{N}\right) \frac{N-n}{N-1} = np(1-p) \frac{N-n}{N-1}.\end{aligned}\quad \square$$

Thus, the expected number of successes in a sample of size n is np regardless of whether we sample with or without replacement. The variance in the case of sampling without replacement is smaller than the corresponding binomial variance by the factor $(N-n)/(N-1)$ (sometimes referred to as *finite population correction factor*).

The behavior of the variances of binomial and hypergeometric distributions is different as the sample size, n , increases. In a binomial distribution, each new element of the sample contributes the same amount to variance, so the latter grows linearly with n . In sampling without replacement, variance changes in proportion to the product $n(N-n)$; hence, it initially grows to reach a maximum when sampling exhausts half of the population, and then declines to zero when $n = N$ (at $n = N$ we exhaust the whole population, so there is no longer variability in the sample). Variance for $n = 1$ is the same as for $n = N - 1$. This is clear, since variability involved in sampling one element is the same as variability involved with leaving just one element unsampled.

The next theorem connects the binomial and hypergeometric distributions in the situation when the successes come from two sources, each following the binomial distribution, and the total number of successes is fixed. Thus, we will prove

Theorem 8.2.3 *Let X and Y be independent random variables with binomial distributions $X \sim \text{BIN}(m, p)$ and $Y \sim \text{BIN}(n, p)$. Then*

$$P\{X = k | X + Y = r\} = \frac{\binom{m}{k} \binom{n}{r-k}}{\binom{m+n}{r}}. \quad (8.26)$$

Proof: Since $X + Y \sim \text{BIN}(m + n, p)$, we have

$$\begin{aligned} P\{X = k | X + Y = r\} &= \frac{P\{X = k, X + Y = r\}}{P\{X + Y = r\}} \\ &= \frac{P\{X = k\}P\{Y = r - k\}}{P\{X + Y = r\}}, \end{aligned}$$

and substitution of binomial probabilities gives (8.26). $\square$

The right-hand side of (8.26) is the hypergeometric probability of k successes in sampling r elements from a population with a total of $m + n$ elements, of which m are successes.

The scheme of sampling without replacement can be generalized as follows: Assume that the population consists initially of a elements of one kind (successes) and $b = N - a$ elements of the second kind (failures). Each time an element is sampled, it is returned, and c elements of the same kind as just sampled are added to the urn. This process continues for n samplings. Let X be the number of successes in the sample. This random variable X is said to have *Pólya distribution*, and the sampling described above is called the *Pólya scheme*.

The reason for designing this scheme was as follows. Adding elements of the same kind to the population increases the probability of selecting elements of the kind most recently sampled. Such effect is known to occur when one samples from a population to determine the fraction of persons infected with a disease. Typically, if one finds one person with the disease, then the chances of finding others with the same disease increase. Pólya introduced this scheme of sampling to model such effects.

Observe that we can formally put $c = -1$ (elements are simply not returned). Thus, the special case $c = -1$ yields random variable X with a hypergeometric distribution. To find $P\{X = k\}$ in the general case, let us first find the probability of sampling the elements in a specific order, for example, first k successes and then $n - k$ failures. This probability, by the chain rule (4.5), is

$$\frac{a}{N} \times \frac{a+c}{N+c} \cdots \frac{a+(k-1)c}{N+(k-1)c} \times \frac{b}{N+kc} \times \frac{b+c}{N+(k+1)c} \cdots \frac{b+(n-k-1)c}{N+(n-1)c}. \quad (8.27)$$

Let us observe that the probability (8.27) remains the same regardless of the order of k successes and $n - k$ failures. Each time the number of elements in the population increases by c , the product of all denominators is $N(N+1) \cdots (N+(n-1)c)$. Similarly, the numerators corresponding to sampling successes are $a, a+c, a+2c, \dots, a+(k-1)c$ regardless of when the successes are to occur, and the same holds for failures. Consequently, we have

Theorem 8.2.4 *In Pólya scheme, for $k = 0, 1, \dots, n$*

$$P\{X = k\} = \frac{n!p(p+\gamma) \cdots (p+(k-1)\gamma)q(q+\gamma) \cdots (q+(n-k-1)\gamma)}{k!(n-k)!(1+\gamma)(1+2\gamma) \cdots (1+(n-1)\gamma)},$$

where $p = a/N, q = b/N = 1 - a/N$ and $\gamma = c/N$.

PROBLEMS

8.2.1 An urn contains nine chips, five of them red and four blue. Three chips are drawn without replacement. Find the distribution of $X =$ number of red chips drawn.

8.2.2 An urn contains six chips, three red and three green. Four chips are selected without replacement. Find $E(X)$ and $\text{Var}(X)$, where $X =$ number of red chips in the sample.

8.2.3 Kim and Jason bought a package of 10 lottery tickets, knowing that each package had to include three winning tickets. Since Kim paid 60% of the total cost, while Jason paid 40%, they took six and four tickets, respectively. Find the probability that: **(i)** Kim got at least one winning ticket; **(ii)** Jason got exactly one winning ticket; and **(iii)** Each of them got at least one winning ticket.

8.2.4 Answer (i)–(iii) in Problem 8.2.3 if instead of three winning tickets in each package, 30% of all tickets are winning and 10 tickets for each package are selected at random.

8.2.5 Instead of (8.23), write $X = \eta_1 + \eta_2 + \cdots + \eta_a$, where $\eta_j = 1$ or 0 depending on whether or not the j th element representing success was selected. Use this representation to derive formulas for the mean and variance of X as given in Theorem 8.2.2.

8.3 POISSON DISTRIBUTION AND POISSON PROCESS

We start from the following definition:

Definition 8.3.1 A random variable X is said to have a Poisson distribution, $\text{POI}(\lambda)$, if for some $\lambda > 0$,

$$P\{X = n\} = \frac{\lambda^n}{n!} e^{-\lambda}, \quad n = 0, 1, \dots \quad (8.28)$$

□

First, we want to check that the terms in (8.28) add to 1. We have

$$\sum_{n=0}^{\infty} P\{X = n\} = e^{-\lambda} \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} = e^{-\lambda} e^{\lambda} = 1.$$

To determine the moments of the Poisson distribution, let us compute the mgf:

$$\begin{aligned} m_X(t) &= Ee^{tX} = \sum_{n=0}^{\infty} e^{nt} P\{X = n\} = \sum_{n=0}^{\infty} e^{nt} \frac{\lambda^n}{n!} e^{-\lambda} \\ &= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda e^t)^n}{n!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}. \end{aligned}$$

Thus, a moment generating function of the Poisson distribution is defined for all t and is differentiable an arbitrary number of times. We have

$$E(X) = m'_X(t)|_{t=0} = \lambda e^t e^{\lambda(e^t - 1)}|_{t=0} = \lambda.$$

An easy differentiation yields $E(X^2) = m''_X(t)|_{t=0} = \lambda^2 + \lambda$, and therefore

$$\text{Var}(X) = \lambda.$$

The next theorem shows that the family of Poisson distributions is closed under addition of independent random variables:

Theorem 8.3.1 *If X and Y are independent, with distributions $\text{POI}(\lambda_1)$ and $\text{POI}(\lambda_2)$, respectively, then $X + Y$ has a $\text{POI}(\lambda_1 + \lambda_2)$ distribution.*

Proof: We will present a direct proof, to show the kind of calculations involved in evaluating the distribution of the sum. The proof using moment generating functions can be found in Example 7.28.

$$\begin{aligned}
 P\{X + Y = n\} &= \sum_{j=0}^n P\{X = j, Y = n - j\} = \sum_{j=0}^n P\{X = j\} P\{Y = n - j\} \\
 &= \sum_{j=0}^n \frac{\lambda_1^j e^{-\lambda_1}}{j!} \times \frac{\lambda_2^{n-j} e^{-\lambda_2}}{(n-j)!} = \frac{e^{-(\lambda_1 + \lambda_2)}}{n!} \sum_{j=0}^n \binom{n}{j} \lambda_1^j \lambda_2^{n-j} \\
 &= \frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^n}{n!} \quad (\text{by Newton's formula}),
 \end{aligned}$$

which completes the proof. □

Finally, we introduce an analogue of Theorem 8.2.3.

Theorem 8.3.2 *If X and Y are independent and $X \sim \text{POI}(\lambda_1)$, $Y \sim \text{POI}(\lambda_2)$, then for $k = 0, 1, \dots, n$,*

$$P\{X = k | X + Y = n\} = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}. \quad (8.29)$$

Thus, the conditional distribution of X given $X + Y$ is binomial, with the number of trials $X + Y$ and the probability of success $p = \lambda_1 / (\lambda_1 + \lambda_2)$.

Proof: Using Theorem 8.3.1, we have

$$\begin{aligned}
 P\{X = k | X + Y = n\} &= \frac{P\{X = k, X + Y = n\}}{P\{X + Y = n\}} \\
 &= \frac{P\{X = k\} P\{Y = n - k\}}{P\{X + Y = n\}} \\
 &= \frac{\frac{\lambda_1^k}{k!} e^{-\lambda_1} \frac{\lambda_2^{n-k}}{(n-k)!} e^{-\lambda_2}}{\frac{(\lambda_1 + \lambda_2)^n}{n!} e^{-(\lambda_1 + \lambda_2)}} = \frac{n!}{k!(n-k)!} \frac{\lambda_1^k \lambda_2^{n-k}}{(\lambda_1 + \lambda_2)^n},
 \end{aligned}$$

which reduces to the right-hand side of (8.29). □

The following theorem explains a very important application of the Poisson distribution:

Theorem 8.3.3 *If $p \rightarrow 0$ and $n \rightarrow \infty$ in such a way that $\lim np = \lambda > 0$, then for $k = 0, 1, \dots$,*

$$\lim_{n \rightarrow \infty} \binom{n}{k} p^k (1-p)^{n-k} = \frac{\lambda^k}{k!} e^{-\lambda}. \quad (8.30)$$

Proof: We will prove (8.30) under simplifying assumption $np = \lambda$ for all n . The proof in the general case is based on the same idea but obscured by some technical points. Replacing p

by λ/n , we have

$$\begin{aligned} \binom{n}{k} p^k (1-p)^{n-k} &= \frac{n(n-1) \cdots (n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right)}{k! \left(1 - \frac{\lambda}{n}\right)^k}. \end{aligned}$$

The factor $(1 - \lambda/n)^n$ converges to $e^{-\lambda}$, while each of the remaining factors involving n converges to 1. Since the number of such factors does not depend on n , their product also tends to 1, which proves the theorem. $\square$

To see the applicability of Theorem 8.3.3, observe that it can be used as an approximation of binomial probability of k successes in n trials, valid for small p and large n :

$$\binom{n}{k} p^k (1-p)^{n-k} \approx \frac{(np)^k}{k!} e^{-np}. \quad (8.31)$$

■ EXAMPLE 8.8

Continuing Example 8.3, we assume that on average, one birth in 80 is a multiple birth. What is the probability that among 400 births that occurred in the maternity ward of a given hospital during the first 3 months of a year there were fewer than 4 multiple births?

SOLUTION. If X stands for the number of multiple births during the period analyzed, then $X \sim \text{BIN}(400, 1/80)$. We have

$$P\{X < 4\} = \sum_{k=0}^3 P\{X = k\} = \sum_{k=0}^3 \binom{400}{k} \left(\frac{1}{80}\right)^k \left(\frac{79}{80}\right)^{400-k}.$$

This sum can be computed directly, and the value is

$$0.0065 + 0.0331 + 0.0835 + 0.1402 = 0.2633.$$

We have here $np = 400 \times \frac{1}{80} = 5$, so the approximation (8.31) by Poisson distribution gives

$$P\{X < 4\} \approx \sum_{k=0}^3 \frac{5^k}{k!} e^{-5} = 0.0067 + 0.0337 + 0.0842 + 0.1404 = 0.2650.$$

The relative errors of consecutive approximating terms are, respectively, 3.08%, 1.31%, 0.34%, and 0.14%, while the relative error of the final answer is 0.55%. Whether or not this may be regarded as a good approximation depends on the goal of finding the probability in question. For most purposes, a relative error below 1% is quite acceptable. One can imagine, however, situations where it need not be so. For instance, an insurance company that is to cover the cost of delivery and hospital care for multiple births might conceivably want to know the probability of fewer than 4 multiple births with precision better than second decimal in order to decide on the premium.

■ EXAMPLE 8.9

Suppose that on average, one in every 100 passengers does not show up for a flight. An airline sold 250 tickets for a flight serviced by an airplane that has 247 seats. What is the probability that every person who shows up for flight will get a seat?

SOLUTION. Let X be the number of passengers who do not show up for the flight in question, and let us treat X as a binomial random variable with $n = 250$, $p = 0.01$, so that $np = 2.5$. The probability of the event $X \geq 3$, using Poisson approximation, is

$$\begin{aligned} P\{X \geq 3\} &= 1 - P\{X < 3\} = 1 - P\{X = 0\} - P\{X = 1\} - P\{X = 2\} \\ &\approx 1 - e^{-2.5} - \frac{2.5}{1!}e^{-2.5} - \frac{(2.5)^2}{2!}e^{-2.5} = 0.4562, \end{aligned}$$

while directly from the binomial distribution we have

$$P\{X \geq 3\} = 1 - \sum_{j=0}^2 \binom{250}{j} (0.01)^j (0.99)^{250-j} = 0.4568. \quad (8.32)$$

This time the approximation by a Poisson distribution to binomial probabilities (8.32) has a relative error of 0.13%.

It has to be pointed out that whereas in Example 8.8 the claim that X has a binomial distribution was fully justified (whether or not a birth is a multiple birth is independent on the multiplicity of other births in the same period), the situation is not so clear in case of passengers missing an airline flight. The point is that people often fly together (typically in families or other groups). In these cases, the fact that one person misses the flight may affect the chances of some other persons missing the same flight. Consequently, X is at best *approximately* binomial. Our calculations therefore give a relative error of an approximation to a number that is already an approximation (to the actual probability).

We will now try to capture features responsible for the fact that the number of occurrences of some event in a given interval of time follows a Poisson distribution.

We consider the class of situations in which a certain event occurs at random points in time. Examples are quite common: arrivals of customers at service stations, twin births in a hospital, earthquakes of specified intensity occurring in a given region, fire alarms in a given town, and so on. To increase practical applicability, the theory focuses only on the times of their occurrence, disregarding other specific features of the events under consideration. The random variable one needs to analyze here is the number $N_{[t_1, t_2]}$ of events that occur between times t_1 and t_2 . The theory built for analyzing such processes in the most general case is called the *theory of point processes*. We will analyze only a special case of point processes, the Poisson process.

The assumptions underlying the Poisson processes attempt to capture the intuitive notion of “complete randomness.” In particular, in the Poisson process knowledge of the past provides no clue in regard to the future.

To express the properties that will imply that a given stream of events is a Poisson process, it will be convenient to introduce a mathematical notation, which will later be useful also in other contexts: The symbol $o(x)$ denotes any function $f(x)$ such that

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = 0.$$

■ EXAMPLE 8.10

A power function x^a is $o(x)$ if $a > 1$, and so is every function of the form $x^a h(x)$ if $a > 1$ and h is continuous at 0 (hence bounded in the neighborhood of 0). For instance, if S_n is a binomial random variable, then, $P\{S_n = k\}/p = \binom{n}{k} p^{k-1} (1-p)^{n-k}$ converges to 0 when $p \rightarrow 0$ if $k > 1$ and so $P\{S_n = k\} = o(p)$ for $k > 1$.

We will often use the following facts:

1. If $\lim_{x \rightarrow 0} \frac{h(x)}{x} = c \neq 0$, then $h(x) = cx + o(x)$.
2. If functions $f_1, f_2, \dots, f_N$ are $o(x)$, then $f_1 + \dots + f_N$ is also $o(x)$.

We can now formulate the postulates of the Poisson process.

Postulate 1. *The numbers of events occurring in two nonoverlapping time intervals are independent.*

Postulate 2. *The probability of at least one event occurring in an interval of length Δt is $\lambda \Delta t + o(\Delta t)$ for some constant $\lambda > 0$.*

Postulate 3. *The probability of two or more events occurring in an interval of length Δt is $o(\Delta t)$.*

The first postulate is the one that asserts that knowledge of the past is of no help in predicting the future. The second postulate asserts stationarity, in the sense that probability of an event occurring in a short time interval is (roughly) proportional to the length of this interval, but does not depend on the location of this interval. Finally, the third postulate asserts that events occur one at a time. That is chances of two events occurring within an interval of length Δt become negligible as $\Delta t \rightarrow 0$.

Let us now fix the zero on time scale, and let $P_n(t)$ denote the probability of exactly n events prior to t , so that $P_n(t) = P\{N_{[0,t)} = n\}$. We will prove

Theorem 8.3.4 *Under all three postulates*

$$P_n(t) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n = 0, 1, \dots \quad (8.33)$$

Proof: By postulates 2 and 3, for every t and $\Delta t > 0$,

$$\begin{aligned} P\{N_{[t, t+\Delta t)} = 1\} &= \lambda \Delta t + o_1(\Delta t), \\ P\{N_{[t, t+\Delta t)} = 0\} &= 1 - \lambda \Delta t + o_2(\Delta t). \end{aligned} \quad (8.34)$$

For $n = 0$ we write, using postulate 1 and (8.34),

$$P_0(t + \Delta t) = P\{N_{[0,t)} = 0, N_{[t, t+\Delta t)} = 0\} = P_0(t)[1 - \lambda \Delta t + o_2(\Delta t)]$$

which gives the difference ratio

$$\frac{P_0(t + \Delta t) - P_0(t)}{\Delta t} = -\lambda P_0(t) + \frac{1}{\Delta t} o_2(\Delta t) P_0(t). \quad (8.35)$$

Passing to the limit with $\Delta t \rightarrow 0$, we obtain¹⁴

$$P'_0(t) = -\lambda P_0(t). \quad (8.36)$$

¹⁴The limit of the right-hand side exists and is equal to the right derivative of P_0 at t . To justify the existence of the left derivative in (8.35), observe that one can replace t by $t - \Delta t$ in (8.35). Since $o_2(\Delta t)$ does not depend on t , we see that P_0 is continuous and that the left derivative is equal to the right derivative.

The initial condition is $P_0(0) = 1$, and the relation of (8.36) gives

$$P_0(t) = e^{-\lambda t}. \quad (8.37)$$

Now, if $n \geq 1$, we write

$$\begin{aligned} P_n(t + \Delta t) &= \sum_{j=0}^n P\{N_{[0,t)} = n - j, N_{[t,t+\Delta t)} = j\} \\ &= \sum_{j=0}^n P_{n-j}(t) P\{N_{[t,t+\Delta t)} = j\} \\ &= P_n(t)[1 - \lambda\Delta t + o_2(\Delta t)] + P_{n-1}(t)[\lambda\Delta t + o_1(t)] + o_3(t), \end{aligned}$$

where $o_3(t)$ is the term obtained from combining together all terms involving two or more events occurring between t and $t + \Delta t$.

Forming the difference ratio and passing to the limit with $\Delta t \rightarrow 0$, we obtain the equations, valid for $n = 1, 2, \dots$,

$$P'_n(t) = -\lambda P_n(t) + \lambda P_{n-1}(t), \quad (8.38)$$

which can be solved recursively using (8.37), with the initial conditions being $P_n(0) = 0, n = 1, 2, \dots$. Alternatively, we could use induction to check that probabilities (8.33) satisfy (8.37) and (8.38). $\square$

We will now discuss some examples of Poisson processes.

■ EXAMPLE 8.11

The maternity ward in a certain hospital has, on average, 30 births per week. Given that there were 6 births on a specific day, find the probability of (a) three births on each of the following two days; (b) a total of six births during the following two days; (c) the expected number of days with exactly one birth during the month of May.

SOLUTION. We assume here that the births in the maternity ward in question form a Poisson process. Consequently, the number of births on a given day does not affect the number of births in future intervals. To answer all these questions, we must first choose the unit of time. This is a totally arbitrary choice, but the important point is that once this choice is made, we must express the parameter λ in the chosen units. Then λ is the expected number of events in the unit of time.

Let a time unit equal one day. Then $\lambda = 30/\text{week} = 4.286/\text{day}$. Consequently, the probability of three births in a given day is $(\lambda^3/3!)e^{-\lambda} = 0.1806$, and probability of such event on two consecutive days is $[(\lambda^3/3!)e^{-\lambda}]^2 = 0.0326$. As regards (b), the probability of 6 births in 2 days is $[(2\lambda)^6/6!]e^{-2\lambda} = 0.1043$. For (c), the number of days in May when there is exactly one birth is has binomial distribution with $n = 31$ and $p = \lambda e^{-\lambda} = 0.0590$, so the expectation equals $31p = 1.83$.

As mentioned, the postulates of Poisson process attempt to capture the idea of “complete randomness.” The theorems below indicate to which extent this attempt is successful.

To facilitate formulation of the theorems, we let $X_t = N_{(0,t)}$ denote the number of events occurring in $(0, t)$, and also let $T_1, T_2, \dots$ denote the times of occurrence of successive events. Thus,

$$T_k = \inf\{t : X_t \geq k\}.$$

In the analogy with (8.19), we have the following identity:

$$T_k \leq t \quad \text{if and only if} \quad X_t \geq k$$

and consequently $P\{T_k \leq t\} = P\{X_t \geq k\}$. Since X_t has Poisson distribution with parameter λt , we have

$$P\{X_t \geq k\} = \sum_{j=k}^{\infty} \frac{(\lambda t)^j}{j!} e^{-\lambda t} = 1 - \sum_{j=0}^{k-1} \frac{(\lambda t)^j}{j!} e^{-\lambda t}.$$

Then the cdf of T_k is

$$P\{T_k \leq t\} = P\{X_t \geq k\} = 1 - \sum_{j=0}^{k-1} \frac{(\lambda t)^j}{j!} e^{-\lambda t}. \quad (8.39)$$

Now, let $U_1 = T_1, U_2 = T_2 - T_1, U_3 = T_3 - T_2, \dots$ be the time to the first event (U_1) and consecutive times between events ($U_2, U_3, \dots$). From (8.39) for $k = 1$ we have $P\{U_1 \leq t\} = P\{T_1 \leq t\} = 1 - e^{-\lambda t}$, so U_1 has an exponential distribution with mean $1/\lambda$. Next

$$\begin{aligned} P\{U_2 > t | T_1 = \tau\} &= P\{U_2 > t | U_1 = \tau\} \\ &= P\{\text{no events in } (\tau, t + \tau) | T_1 = \tau\} \\ &= P\{\text{no events in } (\tau, t + \tau)\} = e^{-\lambda t}, \end{aligned}$$

which means that U_2 also has exponential distribution with mean $1/\lambda$ and is independent of U_1 . Since the argument can be repeated for all other U'_j s, we have proved the following theorem:

Theorem 8.3.5 *In a Poisson process, the time U_1 until the first event and the times $U_2, U_3, \dots$ between subsequent events are independent random variables, each with the same exponential distribution with mean $1/\lambda$.*

Since the origin of time scale $t = 0$ was chosen arbitrarily, this theorem asserts that if we start observing a Poisson process at an arbitrarily selected time, fixed or randomly chosen,¹⁵ then the waiting time for the first event has the same distribution as the times between subsequent events. This property is connected closely with the memoryless property of geometric distribution, specified in Theorem 8.1.2.

Theorem 8.3.6 *If X is a random variable with exponential distribution, then for all $s, t > 0$,*

$$P\{X > s + t | X > s\} = P\{X > t\}. \quad (8.40)$$

Conversely, if X may assume only positive values and satisfies (8.40) for all $s, t > 0$, then X has exponential distribution.

Proof: Let X have exponential distribution. The left-hand side of (8.40) is the ratio

$$\frac{P\{X > s + t\}}{P\{X > s\}} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P\{X > t\}.$$

¹⁵The phrase “randomly chosen” ought to be qualified here. Suppose that the “random choice” is to start observing Poisson process 5 minutes *before* the next event. Technically, such a choice gives a random moment of beginning of observation (since the time of event is random), and for this choice $U_1 = 5$ minutes. Here the qualification of “random choice” is that the decision depends on the past but *not* on the future of the process.

Conversely, (8.40) implies that the tail of the cdf of X , that is, the function $\phi(x) = P\{X > x\}$, satisfies the equation $\phi(s+t) = \phi(s)\phi(t)$. One then shows (e.g., see Feller, 1968) that any bounded solution of this equation must be of the form $\phi(t) = e^{-\lambda t}$ for some $\lambda > 0$. $\square$

The property of a Poisson process discussed above is one of the arguments for the claim that assumptions of a Poisson process capture “maximal randomness.” While in the Poisson process, knowledge of the past does not give a clue to the future; it is not so for other streams of events. For instance, if one arrives at a bus stop just after a bus has left (knowledge of the past), one may expect a longer wait for the next bus.

The following theorem shows that in Poisson processes, knowledge of the number of events in the past gives us, in a sense, no additional information not only about the future but also about the past. We have the following:

Theorem 8.3.7 *The events in Poisson process satisfy the following property: for every $0 < u < t$,*

$$P\{T_1 < u | X_t = 1\} = \frac{u}{t}. \quad (8.41)$$

Proof: We have

$$P\{T_1 < u | X_t = 1\} = \frac{P\{T_1 < u, X_t = 1\}}{P\{X_t = 1\}} = \frac{P\{T_1 < u, X_t = 1\}}{\lambda t e^{-\lambda t}}. \quad (8.42)$$

Conditioning on T_1 , the time of occurrence of the first event, and using the fact that T_1 has exponential distribution, the numerator in (8.42) becomes

$$P\{T_1 < u, X_t = 1\} = \int_0^u P\{X_t = 1 | T_1 = z\} \lambda e^{-\lambda z} dz.$$

If $T_1 = z$, the event $X_t = 1$ occurs if the time U_2 between the first and second event exceeds $t - z$. So

$$P\{X_t = 1 | T_1 = z\} = P\{U_2 > t - z\} = e^{-\lambda(t-z)}.$$

Substituting (8.40), we obtain formula (8.41). $\square$

This theorem asserts that if we know that only one event occurred between 0 and t , then the conditional distribution of the time of occurrence of this event is uniform on $(0, t)$. In a sense, then, we have no information as to when the event occurred.

Finally, let us observe that we can obtain the unconditional density of T_k by differentiating the cdf given by formula (8.39).

Theorem 8.3.8 *The density $f_{T_k}(t)$ of the time T_k of the k th event in Poisson process is*

$$f_{T_k}(t) = \frac{\lambda^k}{(k-1)!} t^{k-1} e^{-\lambda t}, \quad t > 0. \quad (8.43)$$

The distribution of T_k is sometimes called the Erlang distribution. As we will see later, (8.43) is a special case of the gamma density.

■ EXAMPLE 8.12

Fires in a certain town occur according to a Poisson process. If there were 10 fires in a given week, what is the probability that at least one of them occurred on Friday?

SOLUTION. What is of interest here is that we do not need to know the intensity λ of the Poisson process in question. We know that given that the number of fires was 10, their times of occurrence fall within a week according to the uniform distribution. The probability that a single fire does not fall on Friday is $6/7$; hence, chances of at least 1 of 10 fires falling on Friday is $1 - (6/7)^{10} = 0.7859$.

■ EXAMPLE 8.13

Suppose that traffic accidents on a given intersection occur according to a Poisson process, with the rate on Saturdays being twice the rate on weekdays and the rate on Sundays being double the rate on Saturdays. The total rate is about five accidents per week. What is more likely: two accidents on each of two consecutive weekends (Saturday + Sunday), or a total of four accidents on weekdays in a given week?

SOLUTION. If λ is the average number of accidents on a weekday, then it is 2λ on a Saturday and 4λ on a Sunday. Consequently, we have $5\lambda + 2\lambda + 4\lambda = 5$, which gives $\lambda = 5/11$. The number of accidents on a weekend is the sum of the numbers of accidents on Saturday and Sunday. These are independent Poisson random variables; hence, their sum (see Theorem 8.3.1) also has Poisson distribution, with the mean $2\lambda + 4\lambda = 30/11$. Consequently, the first probability is $P\{X = 2\}^2$, where $X \sim \text{POI}(30/11)$, or $[(30/11)^2 e^{-30/11}/2!]^2 = (0.2432)^2 = 0.0592$. The second probability is $P\{Y = 4\}$, where now $Y \sim \text{POI}(5\lambda) = \text{POI}(25/11)$, which equals $(25/11)^4 e^{-25/11}/4! = 0.1145$.

To continue, suppose that on a Friday (which happened to be Friday the 13th) the number of accidents was as high as the number of accidents on the following weekend. Is such an event unusual?

Let us evaluate the chances of such an occurrence without reference to any magic connected with Friday the 13th, that is, probability $P\{X = Y\}$, where X and Y are independent, $X \sim \text{POI}(\lambda)$ and $Y \sim \text{POI}(6\lambda)$ for $\lambda = 5/11$.

$$\begin{aligned} P\{X = Y\} &= \sum_{j=0}^{\infty} P\{X = j\}P\{Y = j\} \\ &= \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} e^{-\lambda} \frac{(6\lambda)^j}{j!} e^{-6\lambda} = \sum_{j=0}^{\infty} \frac{(6\lambda^2)^j}{(j!)^2} e^{-7\lambda} \\ &= 0.0415 + 0.0515 + 0.0159 + 0.0022 + \cdots = 0.1111. \end{aligned}$$

As we see, the chances here are slightly over 10%; hence, such an event need not necessarily be regarded as highly unusual. However, as one may note, almost all of this probability is due to the possibility that $X = Y = 0$ or $X = Y = 1$.

Similarly, the probability that there are more accidents on Friday the 13th than on the whole following weekend is still not negligible:

$$P(X > Y) = \sum_{k=1}^{\infty} P\{X = k\} \sum_{j=0}^{k-1} P\{Y = j\} = 0.0406.$$

This time most of the probability is contributed by the terms $P\{X = 1\} \times P\{Y = 0\}$ and $P\{X = 2\} \times [P\{Y = 0\} + P\{Y = 1\}]$.

The concept of a Poisson process allows a number of generalizations. First of all, note that the symbol t need not be interpreted as time: it may be some other attribute interpretable as a

linear dimension. Thus, one can regard faults on a scotch tape or misprints in a text (regarded as a continuous string of letters) as Poisson processes, provided that one can reasonably expect that the postulates of the Poisson process hold.

One of the generalizations of the Poisson process concerns the extension to a higher dimension. Instead of events occurring in time (i.e., random points on a line), one can consider the case of points allocated at random on a plane or in space. The postulates of the Poisson process in such cases are analogous to the postulates in one dimension. The basic random variable is $X(A)$ = number of points falling into A , where A is a set on the plane or in space. The main postulate asserts that the numbers of points falling into disjoint sets are independent. The second postulate asserts that the probability that $X(A) = 1$ depends on the size of the set A , not on its location, and equals $\lambda|A| + o(|A|)$, where $|A|$ stands for the area or volume of the set A . Finally, the third postulate asserts that the probability that $X(A) \geq 2$ is of the order $o(|A|)$. Under these postulates one can show that

$$P\{X(A) = k\} = \frac{(\lambda|A|)^k}{k!} e^{-\lambda|A|}, \quad k = 0, 1, \dots \quad (8.44)$$

As in a one-dimensional case, λ is the expected number of points falling into a region of unit size.

■ EXAMPLE 8.14

The data below are taken from Feller (1968) who lists them among examples of phenomena fitting the Poisson distribution. The observations concerned points of hits in south London by flying bombs during World War II. The entire area under study was divided into $N = 576$ small areas of 0.25km^2 each, and the numbers N_k of areas that were hit k times were counted. The total number of hits is $T = \sum kN_k = 537$, so the average number of hits per area is $\lambda = T/N = 0.9323$. The fit of the Poisson distribution is excellent, as can be seen from comparison of the actual numbers N_k and expected numbers $NP\{X = k\} = N \frac{\lambda^k}{k!} e^{-\lambda}$ for $\lambda = 0.9323$.

k	0	1	2	3	4	5 or more
N_k	229	211	93	35	7	1
$NP\{X = k\}$	226.74	211.39	98.54	30.62	7.14	1.57

The chi-square goodness-of-fit criterion (to be discussed in further chapters) shows that in about 88% of cases one should expect worse agreement.

This example has become sort of a “classic” in the sense of being reproduced in numerous textbooks on statistics, invariably without any comments (other than remarks that the fit is very good). However, the fit to the Poisson distribution was a piece of information of considerable value as military intelligence: it showed the state of German technology in regard to the precision of their aiming devices. Perfect randomness of hits in a given large area indicated that it was not possible to select any specific target within this area.

Feller writes: “It is interesting to note that most people believed in a tendency of the points of impact to cluster. If this were true, there would be a higher frequency of areas with either many hits or no hit, and a deficiency in the intermediate classes. The data indicates perfect randomness and homogeneity of the area; we have here an instructive illustration of the established fact that, to the untrained eye, randomness appears as regularity or tendency to cluster.” It appears that Feller fully knew the reason for collecting and analyzing the data

in question but could only make a veiled allusion: his book was first published in 1950, just 5 years after the end of World War II, when many things were still secret.

At the end, we will give an example involving the Poisson process in three dimensions.

■ EXAMPLE 8.15

The Poisson Bakery makes a special kind of cookies, called Four-Raisin cookies. Raisins (10,000) are added to the dough for 2,500 cookies, and after thorough mixing, the dough is divided into equal parts of which 2,500 cookies are formed and baked. What is the proportion of Four-Raisin cookies that have no raisins at all? What proportion will have exactly four raisins?

SOLUTION. Here we have a spatial Poisson process, with raisins playing the role of points located randomly in space. If we take a cookie as a unit of volume, then $\lambda = 4$, as there are, on average, four raisins per cookie. If we let X denote the number of raisins in a randomly selected cookie, then $P\{X = 0\} = e^{-4} = 0.0183$, so slightly below 2% of all Four-Raisin cookies are raisin-less. On the other hand, $P\{X = 4\} = (4^4/4!)e^{-4} = 0.1954$.

Now, suppose that you buy a box of 200 Four-Raisin cookies. What is the probability that no more than two of them have no raisins? We have here a situation of a Poisson approximation to the binomial distribution, with “success” being a cookie with no raisins so that $np = 200e^{-4} = 3.66$. The number Y of raisinless cookies in the box has a binomial distribution $\text{BIN}(200, 0.0183)$, which is approximated by the Poisson distribution with $\lambda = np = 3.66$. Therefore, $P\{X \leq 2\} = P\{Y = 0\} + P\{Y = 1\} + P\{Y = 2\} \approx (1 + 3.66 + 3.66^2/2)e^{-3.66} = 0.29$; hence, $P\{Y \geq 3\}$ is about 0.71, which means that 71% of all boxes will contain three or more Four-Raisin cookies with no raisins at all.

PROBLEMS

- 8.3.1** Let X have the $\text{POI}(\lambda)$ distribution. Find: **(i)** the mode of X (i.e., the most likely value of X); **(ii)** $P(X \text{ is even})$. (*Hint:* Write the Taylor expansions for e^λ and $e^{-\lambda}$. Any ideas?)
- 8.3.2** A book with 500 pages contains, on average, 3 misprints per 10 pages. What is the probability that there will be more than one page containing at least three misprints?
- 8.3.3** Accidents in a given plant occur at a rate of 1.5 per month. The numbers of accidents in different months are independent and follow the Poisson distribution. Find the probability of: **(i)** five accidents in a period of five consecutive months; **(ii)** one accident in each of five consecutive months.
- 8.3.4** Suppose that the daily numbers of ships arriving to a certain port are independent, each with $\text{POI}(3)$ distribution. Find: **(i)** the expected number of days in April when there are no arrivals; **(ii)** the expected number and variance of days during the summer months (June, July, August) with the number of arrivals equal to the mean daily arrival rate.
- 8.3.5** A certain store makes, on average, two sales per hour between 9:00 a.m. and 2:00 p.m., and three sales per hour between 2:00 p.m. and 9:00 p.m. The numbers of sales in different time periods are independent and have a Poisson distribution. Find: **(i)** the probability of more than three sales between 10:00 a.m. and noon, and also between

1:00 p.m. and 3:00 p.m. **(ii)** the probability that the number of sales between 10:00 a.m. and 11:00 a.m. will be the same as number of sales between 6:00 p.m. and 7:00 p.m.

- 8.3.6** Weekly numbers of traffic accidents at intersections A, B, and C are independent, each with a Poisson distribution. It is known that, on the average, the number of accidents at intersection A is the same as the number of accidents at intersections B and C combined, while the average number of accidents at intersection B is half of that at intersection C. **(i)** If there were, in a given week, 16 accidents at intersections A, B, and C together, what is the probability that exactly four of them were at intersection C? **(ii)** What is the probability that there were more accidents at intersection C than at intersection A?
- 8.3.7** Find the approximate probability that in 1,000 randomly chosen persons there are exactly: **(i)** Two born on New Year and two born on Christmas. **(ii)** Four born on either Christmas or New Year.
- 8.3.8 (Does Nature Prefer Even Numbers?)** Generalizing Problem 8.3.1 (ii), let X be an integer-valued random variable such that $X = X_1 + X_2$, where X_1, X_2 are independent, identically distributed integer-valued random variables. Show that $P\{X \text{ is even}\} \geq 0.5$ (this property has been pointed out to us by Steve MacEachern, personal communication).
- 8.3.9** Let X be the number of failures preceding the r th success in a sequence of Bernoulli trials with probability of success p . Show that if $q \rightarrow 0, r \rightarrow \infty$ in such a way that $rq = \lambda > 0$, then

$$P\{X = k\} \rightarrow \frac{\lambda^k}{k!} e^{-\lambda}$$

for every $k = 0, 1, 2, \dots$. This shows that the negative binomial distribution can be, for large r and small q , approximated by a Poisson distribution. (*Hint*: Use an argument similar to that in the proof of Theorem 8.3.3.)

- 8.3.10** Suppose that the number of eggs X laid by a bird has a Poisson distribution. Each egg hatches with probability p , independently of what happens to other eggs. Let V_1 and $V_2, V_1 + V_2 = X$, denote the numbers of eggs that hatch, and the number of eggs that do not hatch, respectively. Show that V_1 and V_2 are independent.
- 8.3.11** Traffic accidents at a given intersection occur following a Poisson process. **(i)** Given that 10 accidents occurred in June, what is the probability that the seventh accident occurred before June 10? **(ii)** If it is known that n accidents occurred in April, what is the expected number of accidents that occurred during the second week of that month?
- 8.3.12** Two parts of a document are typed by two typists. Let X and Y be the numbers of typing errors in the two parts of the paper. Assuming that X and Y are independent and have Poisson distributions with parameters λ_1 and λ_2 , respectively, find the probability that: **(i)** The paper (i.e., two combined parts) has at least two typing errors. **(ii)** The total number of typing errors is m . **(iii)** The first part of the paper has k typing errors given that there are n typing errors altogether.
- 8.3.13** Assume that chocolate chips are distributed within a cake according to a Poisson process with parameter λ . A cake is divided into two parts of equal volume (disregard the possibility of cutting through a chocolate chip). Show that the probability that

each part of the cake has the same number of chips is

$$e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda/2)^{2k}}{(k!)^2}.$$

8.3.14 Consider two independent Poisson processes with the same parameter λ . Let $N_i(t)$, $i = 1, 2$ be the number of events in i th process which occurred up to time t , and let U_T be the set of all those times t with $0 \leq t \leq T$ at which $N_1(t) = N_2(t)$. Find $E(U_T)$ given that: **(i)** $N_1(T) = N_2(T) = 2$. **(ii)** $N_1(T) = 2, N_2(T) = 3$. (*Hint:* Use the fact that the sum of two independent Poisson processes is a Poisson process and Theorem 9.8.5.)

8.4 EXPONENTIAL, GAMMA, AND RELATED DISTRIBUTIONS

We have encountered the exponential distribution on a number of occasions as the distribution of times between events in Poisson process in one dimension. To repeat the definition, a random variable X has exponential distribution $\text{EXP}(\lambda)$ if for some $\lambda > 0$ and $x \geq 0$,

$$F(x) = P\{X \leq x\} = 1 - e^{-\lambda x}, \quad x \geq 0,$$

so that for $x \geq 0$ the density of X is $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$. We know that

$$E(X) = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2},$$

and the mgf of X is

$$m_X(s) = \frac{\lambda}{\lambda - s} \quad \text{for } s < \lambda.$$

The hazard function (see Section 5.5) of exponential random variable is constant:

$$h(x) = \frac{f(x)}{1 - F(x)} = \frac{\lambda e^{-\lambda x}}{e^{-\lambda x}} = \lambda,$$

a property closely related to the memoryless property of an exponential distribution asserted in Theorem 8.3.6.

Readers should be aware of the fact that the phrase “exponential distribution with parameter λ ” is ambiguous, since an exponential distribution is sometimes introduced in the form $f(x) = (1/\lambda)e^{-x/\lambda}$, $x > 0$. In this notation, $E(X) = \lambda$ and $\text{Var}(X) = \lambda^2$. Consequently, unless it is clear whether the parameter appears in the numerator or denominator of the exponent, one could use phrases that convey the information about the mean. Thus, “exponential distribution with mean θ ” will have density $(1/\theta)e^{-x/\theta}$, and so on.

Next, the sums of independent and exponentially distributed random variables also appeared in the Poisson process, as times $T_1, T_2, \dots$ of consecutive events. The cdf's of these random variables were obtained using the identity

$$P\{T_r > t\} = P\{X_t < r\} = \sum_{k=0}^{r-1} \frac{(\lambda t)^k}{k!} e^{-\lambda t}.$$

The density of T_r follows now by the differentiation,

$$f_r(t) = \frac{\lambda^r}{(r-1)!} t^{r-1} e^{-\lambda t}, \quad t > 0. \quad (8.45)$$

Since T_r is the sum of r independent waiting times, each with the same exponential distribution, the mgf of T_r exists for $s < \lambda$ and equals

$$m_{T_r}(s) = \left(\frac{\lambda}{\lambda - s} \right)^r.$$

The definition of distribution of T_r as the sum of waiting times involves using an integer value of r . We have a complete analogy with the binomial distribution (count of successes in fixed number of trials) and negative binomial distribution (count of number of trials up to a fixed number of successes), on the one hand, and the Poisson distribution (number of events until fixed time) and distribution ((8.45)) (time till fixed number of events occurs), on the other hand. However, unlikely as in the case of a negative binomial distribution, r need not be an integer, and we can define the class of distributions comprising densities (8.45) as a special case. To this end let us introduce the following definition:

Definition 8.4.1 For $t \geq 0$ we define the *gamma function* as

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx. \quad (8.46)$$

□

We can show that this function is well defined for all $t \geq 0$. Integration by parts gives

$$\Gamma(t) = (t-1)\Gamma(t-1). \quad (8.47)$$

Since $\Gamma(1) = 1$, by induction, for any integer $n \geq 1$, we obtain

$$\Gamma(n) = (n-1)!$$

Consequently, a gamma function can be seen as an extension of the factorial function, which it “fills in” for noninteger values of n .

Let us now show that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Indeed,

$$\begin{aligned} \Gamma\left(\frac{1}{2}\right) &= \int_0^\infty x^{1/2-1} e^{-x} dx = \int_0^\infty x^{-1/2} e^{-x} dx = \int_0^\infty \frac{\sqrt{2}}{y} e^{-y^2/2} y dy \\ &= 2\sqrt{\pi} \int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy = 2\sqrt{\pi} \frac{1}{2} = \sqrt{\pi}, \end{aligned} \quad (8.48)$$

where we used substitution $x = y^2/2$ and the fact that the last integral equals $1/2$ as a half of the integral of the standard normal density.

We can now introduce the following definition:

Definition 8.4.2 A random variable with density of the form

$$f(x) = \begin{cases} Cx^{\alpha-1}e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{for } x < 0 \end{cases}$$

for some $\alpha > 0$ and $\lambda > 0$ is said to have a gamma distribution with *shape* parameter α and *scale* parameter λ , $\text{GAM}(\alpha, \lambda)$. C is the normalizing constant. □

Since

$$C \int_0^\infty x^{\alpha-1} e^{-\lambda x} dx = 1,$$

by substituting $\lambda x = z$, we easily obtain from (8.46) that

$$C = \frac{\lambda^\alpha}{\Gamma(\alpha)}. \quad (8.49)$$

We can now obtain the moments of gamma distribution. That is,

$$\begin{aligned} E(X) &= \int_0^\infty x f(x) dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^\alpha e^{-\lambda x} dx \\ &= \frac{\lambda^\alpha}{\Gamma(\alpha)} \times \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} = \frac{\alpha}{\lambda} \end{aligned} \quad (8.50)$$

in view of (8.47). Similarly

$$\begin{aligned} E(X^2) &= \int_0^\infty x^2 f(x) dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha+1} e^{-\lambda x} dx \\ &= \frac{\lambda^\alpha}{\Gamma(\alpha)} \times \frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} = \frac{\alpha(\alpha+1)}{\lambda^2}, \end{aligned}$$

and

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{\alpha}{\lambda^2}. \quad (8.51)$$

The moment generating function of gamma distribution can be evaluated as follows:

$$\begin{aligned} m(t) &= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty e^{tx} x^{\alpha-1} e^{-\lambda x} dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\lambda-t)x} dx \\ &= \frac{\lambda^\alpha}{\Gamma(\alpha)} \times \frac{\Gamma(\alpha)}{(\lambda-t)^\alpha} = \frac{1}{(1-t/\lambda)^\alpha}, \end{aligned}$$

provided that $t < \lambda$. It is easy to show that the k th ordinary moment is

$$m_k = \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)\lambda^k}. \quad (8.52)$$

The following closure property of gamma distributions is a consequence of Theorem 7.5.2:

Theorem 8.4.1 *If X and Y are independent with $X \sim \text{GAM}(\alpha_1, \lambda)$, $Y \sim \text{GAM}(\alpha_2, \lambda)$, then $X + Y \sim \text{GAM}(\alpha_1 + \alpha_2, \lambda)$.*

Since the waiting time T_k for the k th event in a Poisson process has a distribution $\text{GAM}(k, \lambda)$, Theorem 8.4.1 (for integers α_1 and α_2) expresses the simple fact that the waiting time $T_{\alpha_1+\alpha_2}$ is the sum of two independent waiting times, T_{α_1} and T_{α_2} .

Let us note that exponential distribution with parameter λ is the same as $\text{GAM}(1, \lambda)$. Also let us recall that in Example 5.27 we found the density of the square of standard normal variable [i.e., of $Y = Z^2$, where $Z \sim N(0,1)$]. This density equals $f(y) = \frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2}$, which we recognize as $\text{GAM}(1/2, 1/2)$. The sums of squares of independent standard normal variables appear so often in statistics, that the distribution of such sum bears its own name:

Definition 8.4.3 For integer ν , the distribution $\text{GAM}(\nu/2, 1/2)$ is called the *chi-square* distribution with ν degrees of freedom. A random variable with such distribution is typically denoted by χ_ν^2 . $\square$

The following theorem will be very important in statistical inference; its proof is an immediate consequence of Theorem 8.4.1.

Theorem 8.4.2 *If $Z_1, \dots, Z_n$ are independent, each with standard normal distribution, then*

$$X = Z_1^2 + \dots + Z_n^2$$

has chi-square distribution with n degrees of freedom.

Moreover, if $X_1, \dots, X_k$ are independent, chi-square distributed random variables, $X_i \sim \chi_{\nu_i}^2$, then $Y = X_1 + \dots + X_k$ has chi-square distribution with ν degrees of freedom, where $\nu = \nu_1 + \dots + \nu_k$.

Another property that will be useful in further chapters devoted to statistical inference is provided by the next theorem:

Theorem 8.4.3 *If random variable X has $\text{GAM}(\alpha, \lambda)$ distribution, then $Y = 2\lambda X$ has $\text{GAM}(\alpha, 1/2)$ distribution. If 2α additionally is a positive integer, then Y has $\chi_{2\alpha}^2$ distribution.*

Proof: Let $Y = aX$, where $a > 0$. Then $x = y/a = \psi(y)$ and $\psi'(y) = 1/a$. Consequently, based on (8.4.2),

$$\begin{aligned} f_Y(y) &= f_X(\psi(y)) \times |\psi'(y)| = \frac{\lambda^\alpha}{\Gamma(\alpha)} \left(\frac{y}{a}\right)^{\alpha-1} e^{-\lambda y/a} \times \frac{1}{a} \\ &= \frac{(\lambda/a)^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-(\lambda/a)y}. \end{aligned} \quad (8.53)$$

The density $f_Y(y)$ has a $\text{GAM}(\alpha, \lambda/a)$ distribution, which is the same as $\chi_{2\alpha}^2$, when $a = 2\lambda$ and 2α is a positive integer. □

With wide availability of statistical packages today, the values of cdf as well as percentiles of chi-square distributions for various numbers of degrees of freedom can be easily obtained. For given n and x , we can obtain $P\{\chi_n^2 \leq x\}$, as well as for given n and p , we can obtain x such that $P\{\chi_n^2 \leq x\} = p$.

At the end of this section, we will introduce two families of distributions somewhat related to the exponential family.

Definition 8.4.4 The distribution with the density function

$$f(x; \lambda) = \frac{1}{2} \lambda e^{-\lambda|x|} \quad (8.54)$$

for $-\infty < x < \infty, \lambda > 0$, is called *Laplace* or a *double exponential*. □

The Laplace distribution is symmetric, and it has higher values of a kurtosis than the normal distributions. It can be generalized in several ways. For example, its density could become

$$f(x; \lambda_1, \lambda_2) = \begin{cases} \frac{1}{2} \lambda_1 e^{\lambda_1 x} & \text{for } x < 0 \\ \frac{1}{2} \lambda_2 e^{-\lambda_2 x} & \text{for } x \geq 0, \end{cases}$$

or even

$$f(x; p, \lambda_1, \lambda_2) = \begin{cases} p \lambda_1 e^{\lambda_1 x} & \text{for } x < 0 \\ (1-p) \lambda_2 e^{-\lambda_2 x} & \text{for } x \geq 0, \end{cases} \quad (8.55)$$

where $\lambda_1 > 0$, $\lambda_2 > 0$ and $0 < p < 1$. Such generalized distribution will be skewed if $\lambda_1 \neq \lambda_2$ or $p \neq 0.5$. The Laplace distribution is very useful for modeling in biological sciences, economics, and finance. Interested readers can find more information in Kotz et al. (2001).

The moment generating function of the Laplace distribution ((8.54)) is

$$\begin{aligned} E(e^{tX}) &= \int_{-\infty}^{+\infty} e^{tx} \frac{1}{2} \lambda e^{-\lambda|x|} dx = \frac{\lambda}{2} \int_{-\infty}^0 e^{x(t+\lambda)} dx + \frac{\lambda}{2} \int_0^{+\infty} e^{x(t-\lambda)} dx \\ &= \frac{\lambda e^{x(t+\lambda)}}{2(t+\lambda)} \Big|_{-\infty}^0 + \frac{\lambda e^{x(t-\lambda)}}{2(t-\lambda)} \Big|_0^{+\infty} \\ &= \frac{\lambda}{2} \left(\frac{1}{t+\lambda} - \frac{1}{t-\lambda} \right) = \frac{\lambda^2}{\lambda^2 - t^2}. \end{aligned} \quad (8.56)$$

The density (8.54) of the Laplace distribution is symmetric around 0; therefore, its expected value is 0. The computation of other moments will be left as an exercise.

The last family of distributions to be introduced in this section is named after Waloddi Weibull, a Swedish engineer, scientist, and mathematician. The family of distributions that he introduced has a lot of flexibility and, as such, is widely used in industrial and engineering applications, such as reliability analysis, determination of wind spread distribution, or to model the dispersion of the received signals in radar systems.

Definition 8.4.5 The random variable with density of the form

$$f(x; k, \theta) = k\theta(x\theta)^{k-1} e^{-(x\theta)^k} \quad (8.57)$$

for $x > 0$, $\theta > 0$, $k > 0$ is said to have a *Weibull* distribution, denoted $\text{WEI}(k, \theta)$; k and θ are called its *shape* and *scale* parameters, respectively. $\square$

The cdf of a Weibull distribution is

$$F(x) = 1 - e^{-(x\theta)^k},$$

and consequently, its hazard function equals $k\theta(x\theta)^{k-1}$. For $k < 1$, the hazard function is decreasing, and it is increasing for $k > 1$. If $k = 1$, the hazard function is constant and $\text{WEI}(1, \theta)$ distribution becomes exponential.

While the moment generating function of Weibull distribution is not tractable, it can be shown (we leave this for the readers as an exercise) that ordinary moments, m_n , equal

$$m_n = \theta^{-n} \Gamma\left(1 + \frac{n}{k}\right) \quad (8.58)$$

and that the mean and the variance are $\theta^{-1} \Gamma(1 + \frac{1}{k})$ and $\theta^{-2} [\Gamma(1 + \frac{2}{k}) - \Gamma^2(1 + \frac{1}{k})]$, respectively.

PROBLEMS

8.4.1 Show that if variable X has a $\text{GAM}(n, 1)$ distribution, where n is a positive integer, then its cdf is given by the following formula:

$$F_X(x) = \frac{1}{\Gamma(n)} \int_0^x t^{n-1} e^{-t} dt = 1 - e^{-x} \sum_{j=0}^{n-1} \frac{x^j}{j!}.$$

(Hint: Integrate by parts and use induction.)

- 8.4.2** A system consists of five components. Suppose that the lifetimes of the components are independent, with exponential distributions $\text{EXP}(\lambda_1), \dots, \text{EXP}(\lambda_5)$. Find the cdf and density of variable $T = \text{time to failure of the system}$ if the components are connected: **(i)** In series (see Figure 8.2), so that the system fails as soon as one of its components fails. **(ii)** In parallel (see Figure 8.3), so that the system works as long as at least one component is operating. **(iii)** As in Figure 8.4.



Figure 8.2 Series system.

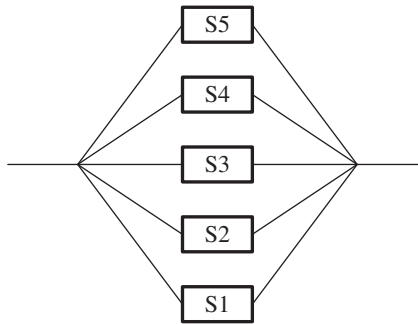


Figure 8.3 Parallel system.

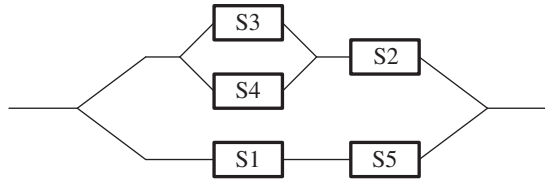


Figure 8.4 Series-parallel system.

- 8.4.3** In the flowchart of Figure 8.5, the block denoted “sample U ” means that the computer samples a value of random variable U with a distribution uniform on $(0, 1)$, the samplings being independent each time the program executes this instruction. Assume that m is a positive integer and $\lambda > 0$. Find: **(i)** $P\{T \leq 2\}$ if $\lambda = 2$ and $m = 3$. **(ii)** The cdf and density of T , $E(T)$, and $\text{Var}(T)$ in the general case.
- 8.4.4** Suppose that patients arrive to the emergency room according to Poisson distribution with mean $\lambda = 3$. What is the expected time until the 9th patient arrives? Find the probability that the next (10th) patient arrives 1 hour later.
- 8.4.5** For a variable X that has a Laplace distribution with $\lambda = 1$, find: **(i)** Interquartile range (the difference between the upper and lower quartile). **(ii)** Variance. **(iii)** Kurtosis.
- 8.4.6** For a variable X having Laplace distribution with density $f(x, 0.4, 1, 2)$ given by (8.55), find: **(i)** Mean; **(ii)** Variance; **(iii)** Median; **(iv)** Interquartile range.

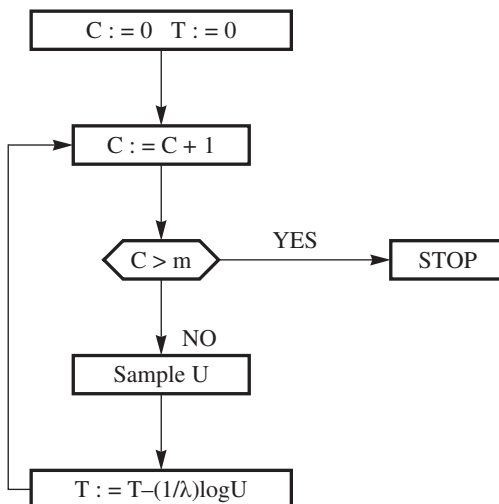


Figure 8.5 Flowchart.

8.4.7 Find the distribution of $X = \theta(-\log U)^{1/k}$, if $U \sim U(0, 1)$.

8.4.8 (i) Find the median and the mode of the Weibull distribution with a density (8.57). **(ii)** Prove the formula (8.58).

8.4.9 It was found that the survival time (in years) in a group of patients who went through a certain medical treatment and are in the similar risk group follows WEI(2, 1/3) distribution. Find: **(i)** The median survival time for such patients. **(ii)** The probability that a randomly selected patient will live at least five more years if he already survived 1 year after the treatment.

8.5 NORMAL DISTRIBUTION

We have already encountered the normal distribution in Chapter 5. Let us recall that the univariate normal distribution with parameters μ and σ^2 , denoted $N(\mu, \sigma^2)$, has density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}, \quad -\infty < x < \infty.$$

It has been shown in Example 5.17 that f is indeed a density. Moreover, if X has distribution $N(\mu, \sigma^2)$, then $E(X) = \mu$, $\text{Var}(X) = \sigma^2$, which gives a direct interpretation of the parameters.

The moment generating function of the normal distribution was obtained in Example 7.27 as

$$m(t) = e^{\mu t + \sigma^2 t^2 / 2}.$$

We have the following closure properties of the normal distribution:

Theorem 8.5.1 *A linear transformation of a normally distributed random variable again has a normal distribution. In particular, if random variable X has a normal distribution $N(\mu, \sigma^2)$, then the random variable*

$$Z = \frac{X - \mu}{\sigma}$$

has the standard normal distribution $N(0, 1)$.

Proof: We have, using (7.21),

$$m_Z(t) = m_{1/\sigma X - (\mu/\sigma)}(t) = m_X(t/\sigma) e^{-\mu/\sigma t} = e^{t^2/2}.$$

The right-hand side is the mgf of a standard normal random variable. $\square$

The property asserted in Theorem 8.5.1 has important practical consequences. It shows that to determine probabilities for *any* normally distributed random variable, it suffices to have access to the probabilities for a standard normal random variable.

This property is especially important in view of the fact that the cdf of a normal distribution

$$\Phi(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-(t-\mu)^2/2\sigma^2} dt$$

cannot be integrated in closed form. Thus, it is necessary to use the tables, and Theorem 8.5.1 implies that one table is sufficient to calculate probabilities for all normal distributions. One can also use any statistical software.

We will let Z denote the standard normal random variable, with density

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

and cdf

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-x^2/2} dx. \quad (8.59)$$

Theorem 8.5.1 asserts, in effect, that given the tables of function (8.59), we have, for any $X \sim N(\mu, \sigma^2)$,

$$\begin{aligned} P\{a \leq X \leq b\} &= P\left\{\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}\right\} \\ &= \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right). \end{aligned}$$

Since $\phi(x)$ is symmetric around 0, we have $P\{Z \leq z\} = P\{Z \geq -z\}$; hence, the cdf of standard normal distribution satisfies the relation

$$\Phi(z) = 1 - \Phi(-z).$$

Consequently, many statistical tables give the values of $\Phi(z)$ only for $z \geq 0$.

Using any statistical software we find $\Phi(3) = 0.9987$. Since for $z > 0$ we have

$$\begin{aligned} P\{|Z| > z\} &= P\{Z > z\} + P\{Z < -z\} \\ &= 1 - \Phi(z) + \Phi(-z) = 2\Phi(-z) = 2[1 - \Phi(z)], \end{aligned}$$

we see that $P\{|Z| > 3\} = 0.0026$. Using Theorem 8.5.1, we see that if $X \sim N(\mu, \sigma^2)$, then

$$P\{|X - \mu| > 3\sigma\} = P\{|Z| > 3\} = 0.0026. \quad (8.60)$$

This explains the origin of the *three-sigma rule*, according to which one is allowed to disregard the possibility of a random variable deviating from its mean more than three standard deviations.

Most tables of standard normal distribution do not give the values of $\Phi(z)$ for $z > 3$. If such values are needed, we have

Theorem 8.5.2 For $z > 0$ the function $\Phi(z)$ satisfies the inequality

$$\left(\frac{1}{z} - \frac{1}{z^3}\right) \phi(z) \leq 1 - \Phi(z) \leq \frac{1}{z} \phi(z) \quad (8.61)$$

and consequently, as $z \rightarrow \infty$,

$$1 - \Phi(z) \approx \frac{1}{z} \phi(z). \quad (8.62)$$

Proof: We have here the obvious inequality

$$\left(1 - \frac{3}{x^4}\right) \phi(x) \leq \phi(x) \leq \left(1 + \frac{1}{x^2}\right) \phi(x).$$

A simple check shows that this inequality is equivalent to

$$-\frac{d}{dx} \left[\left(\frac{1}{x} - \frac{1}{x^2}\right) \phi(x) \right] \leq -\frac{d}{dx} [1 - \Phi(x)] \leq -\frac{d}{dx} \left[\frac{1}{x} \phi(x) \right]. \quad (8.63)$$

Integrating (8.63) between z and ∞ , we obtain (8.61). $\square$

Numerically, the relation (8.62) gives $1 - \Phi(4) \approx 3.36 \times 10^{-5}$, $1 - \Phi(5) \approx 2.97 \times 10^{-7}$, $1 - \Phi(6) \approx 1.01 \times 10^{-9}$. Such small probabilities could be of interest in estimating the chances of, say, an accident in a nuclear power plant.

From a practical viewpoint, the knowledge of these probabilities allows us to assume the normality of distribution in many cases where “logically” the distribution cannot possibly be normal. To illustrate the point, it is often assumed that an attribute as, say, height in human population is normally distributed (e.g., among men, we have mean μ about 70 inches and standard deviation σ of about 2 inches). But since a normally distributed random variable may always assume both positive and negative values, one could argue that height—which cannot be negative—cannot have a normal distribution. The chances of a random variable X with normal distribution $N(70, 4)$ being negative are of the order $\Phi(-35) = 1 - \Phi(35) \approx 0.0069 \times e^{-612}$. So events with such a probability can be safely disregarded, and it turns out that it is much easier to work with the assumption of normality than invent a distribution for height that does not allow negative values.

We will now prove a very useful and important property of normal distribution and illustrate its use.

Theorem 8.5.3 Let X and Y be independent random variables with distributions $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ and let α, β be any constants. Then the random variable $U = \alpha X + \beta Y$ has the distribution $N(\alpha\mu_1 + \beta\mu_2, \alpha^2\sigma_1^2 + \beta^2\sigma_2^2)$. Moreover, if $X_1, \dots, X_k$ are independent random variables with distributions $N(\mu_i, \sigma_i^2)$ for $i = 1, \dots, k$ and if $\alpha_1, \dots, \alpha_k$ are any constants, then random variable $W = \sum_{i=1}^k \alpha_i X_i$ has the $N(\sum_{i=1}^k \alpha_i \mu_i, \sum_{i=1}^k \alpha_i^2 \sigma_i^2)$ distribution.

Proof: We will provide the proof for $k = 2$, the proof for $k > 2$ is analogous. Since $m_X(t) = \exp\{\mu_1 t + \sigma_1^2 \frac{t^2}{2}\}$, $m_Y(t) = \exp\{\mu_2 t + \sigma_2^2 \frac{t^2}{2}\}$, and $\alpha X, \beta Y$ are independent, we have

$$\begin{aligned} m_U(t) &= m_{\alpha X + \beta Y}(t) = m_{\alpha X}(t) m_{\beta Y}(t) = m_X(\alpha t) m_Y(\beta t) \\ &= \exp\left\{\mu_1 \alpha t + \sigma_1^2 \frac{\alpha^2 t^2}{2}\right\} \exp\left\{\mu_2 \alpha t + \sigma_2^2 \frac{\beta^2 t^2}{2}\right\} \\ &= \exp\left\{(\mu_1 \alpha + \mu_2 \beta)t + (\sigma_1^2 \alpha^2 + \sigma_2^2 \beta^2) \frac{t^2}{2}\right\}. \end{aligned}$$

We recognize the right-hand side as an mgf of $N(\mu_1\alpha + \mu_2\beta, \sigma_1^2\alpha^2 + \sigma_2^2\beta^2)$ distribution. $\square$

EXAMPLE 8.16

Assume that the height of men in a certain population is normal with mean $\mu_M = 70$ inches and standard deviation $\sigma_M = 2$ inches. The height of women is also normal, with mean $\mu_W = 68$ inches and $\sigma_W = 1.5$ inches.

One man and one woman are selected at random. What is the probability that the woman is taller than the man?

SOLUTION. Let X and Y be the heights of the randomly selected man (X) and woman (Y). We need $P\{Y > X\}$.

Without Theorem 8.5.3, we could proceed as follows (this solution is applicable to *any* distribution of X and Y , and is therefore of some general interest): If F and G are cdf's for X and Y , respectively, and f and g are their densities, then conditioning on values of Y , we have

$$\begin{aligned} P\{Y > X\} &= \int P\{Y > X | Y = y\} g(y) dy = \int P\{X < y\} g(y) dy \\ &= \int F(y) g(y) dy. \end{aligned} \quad (8.64)$$

We also condition on values of X , obtaining

$$\begin{aligned} P\{Y > X\} &= \int P\{Y > X | X = x\} f(x) dx = \int P\{Y > x\} f(x) dx \\ &= \int [1 - G(x)] f(x) dx. \end{aligned}$$

Using now the assumption of normality of X and Y , based on (8.64), we have

$$\begin{aligned} P\{Y > X\} &= \int_{-\infty}^{+\infty} \left[\int_{-\infty}^y \frac{1}{\sigma_W \sqrt{2\pi}} e^{-(x-\mu_W)^2/2\sigma_W^2} dx \right] \\ &\quad \times \frac{1}{\sigma_M \sqrt{2\pi}} e^{-(y-\mu_M)^2/2\sigma_M^2} dy \\ &= \frac{1}{2(3/2)2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^y e^{-(x-70)^2/8 - (y-68)^2/(18/4)} dx dy. \end{aligned}$$

However, by Theorem 8.5.3, we have

$$P\{Y > X\} = P\{Y - X > 0\} = P\{U > 0\},$$

where $U = Y - X$ has normal distribution with mean $\mu_U = \mu_W - \mu_M = 68 - 70 = -2$ and variance $\sigma_U^2 = \sigma_W^2 + \sigma_M^2 = (3/2)^2 + 2^2 = 25/4$, hence, $\sigma_U = 2.5$ inches (we use here Theorem 8.5.3 with $\alpha = -1, \beta = 1$). Consequently, since U is normally distributed,

$$\begin{aligned} P\{Y > X\} &= P\left\{Z > \frac{0 - (-2)}{2.5}\right\} = P\{Z > 0.8\} \\ &= 1 - \Phi(0.8) = 1 - 0.7881 = 0.2119. \end{aligned}$$

Before we discuss the multivariate normal distribution, we will introduce two distributions that are related to normal: *lognormal* and *folded normal* distributions.

Definition 8.5.1 Random variable X is said to have a *lognormal distribution* if the variable $Y = \log X$ has a normal distribution. The density of a random variable X is then

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma x} \exp \left\{ -\frac{[\log(x) - \mu]^2}{2\sigma^2} \right\},$$

where $N(\mu, \sigma^2)$ is the distribution of Y . □

The mean and the variance of the lognormal distribution are

$$E(X) = e^{\mu + \sigma^2/2} \quad \text{and} \quad V(X) = e^{2\mu + \sigma^2} \times (e^{\sigma^2} - 1),$$

and the general formula for m_k , the k th ordinary moment, which we give without proof, is

$$m_k = \exp \left\{ k\mu + \frac{k^2\sigma^2}{2} \right\}.$$

The lognormal distribution has one very unique property: Even though it has finite moments of all orders, the moment generating function is infinite at any positive number.

The lognormal distribution is widely applicable in modeling, for example, where there is a multiplicative effect of several small independent factors. A typical example is the long-term return rate on a stock investment that can be considered as the product of the daily return rates.

In many experiments, measurements such as dimensions (as well as time and angles) are often recorded with respect to certain expected standards, where the magnitude of the difference, not the direction, is important. Consequently, the actual distribution of the measurement, X , is replaced by a distribution of absolute measurements, $Y = |X|$. When the underlying distribution is normal, the resulting distribution is called the *folded normal*, or *half-normal*.

Any normal distribution $N(\mu, \sigma^2)$ can be “folded” at zero, but we will discuss here only the simplest case with $\mu = 0$. So for $X \sim N(0, \sigma^2)$, the density of $Y = |X|$ is

$$f(y; \sigma) = \sqrt{\frac{2}{\pi}} \times \frac{1}{\sigma} e^{-y^2/2\sigma^2}$$

for $y > 0$, and 0 otherwise. We will now find the expected value $E(Y)$ and the variance $\text{Var}(Y)$. Notice that σ^2 is no longer a variance of the distribution.

$$E(Y) = \int_0^{+\infty} y \sqrt{\frac{2}{\pi}} \times \frac{1}{\sigma} e^{-y^2/(2\sigma^2)} dy = \sigma \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-t} dt = \sigma \sqrt{\frac{2}{\pi}},$$

where we used substitution $t = y^2/2\sigma^2$. For $\text{Var}(Y)$ we need

$$\begin{aligned} E(Y^2) &= \int_0^{+\infty} y^2 \sqrt{\frac{2}{\pi}} \times \frac{1}{\sigma} e^{-y^2/(2\sigma^2)} dy = \frac{1}{2} \int_{-\infty}^{+\infty} x^2 \sqrt{\frac{2}{\pi}} \times \frac{1}{\sigma} e^{-x^2/(2\sigma^2)} dy \\ &= \int_{-\infty}^{+\infty} x^2 \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)} dy = E(X^2) = \sigma^2, \end{aligned}$$

since $X \sim N(0, \sigma^2)$. Finally, we obtain

$$\text{Var}(Y) = \sigma^2 - \left(\sigma \frac{2}{\pi} \right)^2 = \sigma^2 \left(1 - \frac{2}{\pi} \right).$$

Derivation of the moment generating function will be left to the reader as an exercise.

Let us now consider the multivariate normal distribution. We will start from the case of two dimensions. Some situations with more than two dimensions will be considered in later chapters.

Definition 8.5.2 The pair (X, Y) of random variables is said to have a bivariate normal distribution, $N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, if the joint density is of the form

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho \frac{x-\mu_1}{\sigma_1} \times \frac{y-\mu_2}{\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2} \right]}, \quad (8.65)$$

where $-\infty < \mu_1 < \infty, -\infty < \mu_2 < +\infty, \sigma_1 > 0, \sigma_2 > 0$, and $|\rho| < 1$. □

To find marginal distributions of X and Y and determine the interpretation of parameters, let us change the variables into

$$U = \frac{X - \mu_1}{\sigma_1}, \quad V = \frac{Y - \mu_2}{\sigma_2}$$

This amounts to substitution

$$x = \sigma_1 u + \mu_1, \quad y = \sigma_2 v + \mu_2$$

with the Jacobian $J = \sigma_1 \sigma_2$. Consequently, the joint density of (U, V) is

$$g(u, v) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} [u^2 - 2\rho uv + v^2] \right\}.$$

Since $u^2 - 2\rho uv + v^2 = (u - \rho v)^2 + (1 - \rho^2) v^2$, we write

$$g(u, v) = \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2} \left(\frac{u - \rho v}{\sqrt{1-\rho^2}} \right)^2 \right\} \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{v^2}{2} \right\},$$

where

$$\frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} \exp \left\{ -\frac{1}{2} \left(\frac{u - \rho v}{\sqrt{1-\rho^2}} \right)^2 \right\} du = 1$$

for every v . Hence, the second factor is the marginal density of V , while the first factor is the conditional density of U given $V = v$. Thus, V is standard normal, while $U|V \sim N(\rho V, 1 - \rho^2)$. By symmetry, U must also be standard normal and $V|U \sim N(\rho U, 1 - \rho^2)$. Therefore, we have proved

Theorem 8.5.4 If (X, Y) have bivariate normal distribution given by (8.65), then both X and Y have normal distributions, $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively. Moreover, the conditional distributions are also normal:

$$X|Y \sim N \left(\rho \frac{\sigma_1}{\sigma_2} (Y - \mu_2) + \mu_1, \sigma_1^2 (1 - \rho^2) \right) \quad (8.66)$$

and

$$Y|X \sim N \left(\rho \frac{\sigma_2}{\sigma_1} (X - \mu_1) + \mu_2, \sigma_2^2 (1 - \rho^2) \right). \quad (8.67)$$

Proof: Only the last two statements require some proof. The normality of the conditional distribution is obvious in view of the normality of conditional distributions of U given V . We have

$$\begin{aligned} E(X|Y) &= E(\sigma_1 U + \mu_1 | \sigma_2 V + \mu_2) = \sigma_1 E(U | \sigma_2 V + \mu_2) + \mu_1 \\ &= \sigma_1 E(U | V) + \mu_1 = \sigma_1 \rho V + \mu_1 = \sigma_1 \rho \frac{Y - \mu_2}{\sigma_2} + \mu_1. \end{aligned}$$

Similarly,

$$\text{Var}(X|Y) = \text{Var}(\sigma_1 U + \mu_1 | \sigma_2 V + \mu_2) = \sigma_1^2 \text{Var}(U | V) = \sigma_1^2 (1 - \rho^2),$$

which proves (8.66). The proof of (8.67) is analogous. $\square$

Theorem 8.5.5 *If X, Y have the bivariate normal distribution (8.65), then ρ is the coefficient of the correlation between X and Y .*

Proof: Clearly, $\rho_{X,Y} = \text{Cov}(U, V) = E(UV)$, and we may write

$$\begin{aligned} E(UV) &= \int \int uv g(u, v) du dv \\ &= \frac{1}{2\pi\sqrt{1-\rho^2}} \int \int uv \exp \left\{ -\frac{1}{2(1-\rho^2)} [u^2 - 2\rho uv + v^2] \right\} du dv \\ &= \int \frac{ve^{-v^2/2}}{\sqrt{2\pi}} \left[\frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \int u \exp \left\{ -\frac{1}{2} \left(\frac{u-\rho v}{\sqrt{1-\rho^2}} \right)^2 \right\} du \right] dv \\ &= \frac{1}{\sqrt{2\pi}} \int ve^{-v^2/2} \rho v dv = \rho \frac{1}{\sqrt{2\pi}} \int v^2 \exp \left\{ -\frac{v^2}{2} \right\} dv = \rho. \end{aligned}$$

$\square$

■ EXAMPLE 8.17

Assume, as in Example 8.16, that the height of men X and the height of women Y in a population have $N(70, 4)$ and $N(68, 2.25)$ distributions, respectively. Assume also that the heights of siblings are correlated with $\rho = 0.6$. If we sample a brother and a sister, what is the probability that the sister is taller than her brother?

SOLUTION. Proceeding as in Example 8.16, $P\{Y > X\} = P\{Y - X > 0\}$, and we know that $Y - X$ is normal, with $E(Y - X) = E(Y) - E(X) = 68 - 70 = -2$ inches. For the variance, we have

$$\begin{aligned} \text{Var}(Y - X) &= \text{Var}(Y) - 2\text{Cov}(X, Y) + \text{Var}(X) = \sigma_Y^2 - 2\rho\sigma_X\sigma_Y + \sigma_X^2 \\ &= (1.5)^2 - 2 \times 0.6 \times 2 \times 1.5 + 2^2 = 2.65. \end{aligned}$$

Consequently,

$$P\{Y > X\} = P \left\{ Z > \frac{0 - (-2)}{\sqrt{2.65}} \right\} = 1 - \Phi(1.23) = 0.1093.$$

To develop the intuition concerning the bivariate normal distribution and correlation coefficient, let us consider some possible schemes that lead to the appearance of bivariate normal distribution with correlated variables.

■ EXAMPLE 8.18 Sequential Formation

It is possible that one of the values, X , is formed at random, following the normal distribution $N(\mu_1, \sigma_1^2)$. This means that some random process leads to the creation of an element of the population with a specific value x of an attribute X . Subsequently, the value of attribute Y is formed by some other random process that generates Y according to the normal distribution with mean $\rho(\sigma_2/\sigma_1)(x - \mu_1) + \mu_2$ and standard deviation $\sigma_2\sqrt{1 - \rho^2}$.

Examples of such “sequential” generation of attributes are quite common: We could think here of X and Y being the temperatures at noon at a specific place today and tomorrow, water levels on the same river at two specific times, or in two places, one downstream from another, the height of a father and a son, and so on. In a sense, we have a “natural” ordering, with X being the first variable, whose value affects the second variable Y .

To enable a clear geometrical representation, let $\sigma_1 = \sigma_2 = 1$ and let μ_1, μ_2 be sufficiently far from the origin, to have clear plots of marginals, say $\mu_1 = \mu_2 = 5$ (see Figure 8.6).

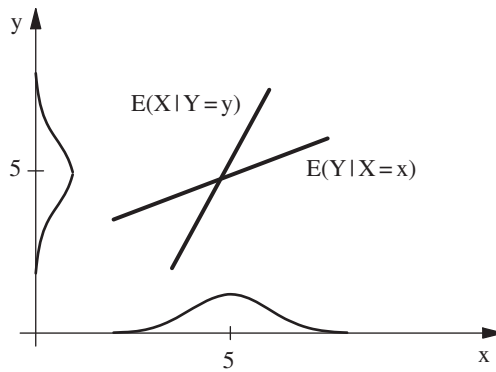


Figure 8.6 Two regression lines.

The conditional expectation of Y , given X (the regression of Y on X , discussed in detail in Chapter 14), is now, by formula (8.67),

$$E(Y|X) = \rho(X - 5) + 5,$$

which is the line $y = \rho x + 5(1 - \rho)$ with slope ρ , passing through the point $(\mu_1, \mu_2) = (5, 5)$. Observe that regression of X on Y is the line $E(X|Y) = \rho(Y - 5) + 5$; that is, $y = (1/\rho)x - 5[(1 - \rho)/\rho]$, which also passes through the point $(5, 5)$ but has the slope $1/\rho$.

After the value of $X = x$ is sampled, the value of Y is sampled from the normal distribution centered at the appropriate point on the regression line, and having variance $1 - \rho^2$. Remembering that the total variance of Y is 1, we have the decomposition of $\text{Var}(Y)$ into a sum of the form $\rho^2 + (1 - \rho^2)$, the second term being the variance of the deviation $Y - [\rho X - 5(1 - \rho)]$ of Y from the regression line. The first term, ρ^2 , is therefore the contribution to the variability of Y coming from the variability of X . In generally accepted terminology, $100\rho^2$ is the “percentage of variance of Y explained by the variability of X .”

A glance at Figure 8.6 reveals that if ρ were larger, the line would be more steep, and this would increase the contribution of X to the variability of Y . (Remember that X and Y are standardized, so their variances remain equal to 1. This explains why the regression line $E(Y|X)$ —in the case of standardized variables—cannot be steeper than the diagonal. So, when ρ increases, to keep $\text{Var}(Y) = 1$, one has to decrease the variance of deviations from the regression line.)

■ EXAMPLE 8.19

Another interpretation of the correlation coefficient ρ is related to the following situation: Suppose that we have a population of objects of some kind and that an attribute ξ of objects in this population has a normal distribution $N(\mu, \sigma^2)$. Elements of this population are sampled and their attribute ξ is measured twice. The measurements are subject to error, and the errors ϵ_1 and ϵ_2 of the two measurements are independent of ξ and from one another, with the same normal distribution $\epsilon_1 \sim N(0, \sigma_\epsilon^2)$, $\epsilon_2 \sim N(0, \sigma_\epsilon^2)$. The observed results of measurement are $X = \xi + \epsilon_1$, $Y = \xi + \epsilon_2$. In this case X and Y are normal with the same means and variances $\mu_1 = \mu_2 = \mu$, $\sigma_1^2 = \sigma_2^2 = \sigma^2 + \sigma_\epsilon^2$. The covariance between X and Y equals $E(XY) - \mu^2 = E(\xi + \epsilon_1)(\xi + \epsilon_2) - \mu^2 = \sigma^2$, in view of the assumed independence and $E(\epsilon_i) = 0$, we have

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\sigma_1^2 \sigma_2^2}} = \frac{\sigma_\xi^2}{\sigma_\xi^2 + \sigma_\epsilon^2}.$$

Here $\rho > 0$, which means that the results of measurements of the same (random) quantity, subject to independent errors, are *always positively correlated*.

It should be pointed out here that—as opposed to Example 8.18— ρ (not ρ^2) represents the fraction of variance of X (or Y) “explained” by the variability of ξ . The situation is different from that in Example 8.18, since now we “explain” the variance of one variable (X), not through the variability of the second variable of the pair (Y), but through the variability of some other variable (ξ) that affects both X and Y .

PROBLEMS

In all problems of this section, Z stands for a standard normal variable, and Φ is its cdf.

8.5.1 Use the tables of normal distribution or any statistical software to determine the probabilities: **(i)** $P(0 \leq Z \leq 1.34)$. **(ii)** $P(0.14 \leq Z \leq 2.01)$. **(iii)** $P(-0.21 \leq Z \leq -0.04)$. **(iv)** $P(-0.87 \leq Z \leq 1.14)$. **(v)** $P(|Z| \geq 1.02)$. **(vi)** $P(Z \geq 1.11)$.

8.5.2 Determine x in the following cases: **(i)** $\Phi(x) = 0.62$. **(ii)** $\Phi(x) = 0.45$. **(iii)** $P(|Z| \leq x) = 0.98$. **(iv)** $P(1.4 \leq Z \leq x) = 0.12$.

8.5.3 Find $P(|X - 2| \leq 0.5)$ if $X \sim N(1, 4)$.

8.5.4 Let random variable X have a $N(\mu, \sigma^2)$ distribution. Find: **(i)** μ if $\sigma^2 = 2$ and $P(X \leq 12) = 0.72$. **(ii)** σ^2 if $\mu = 2$ and $P(X \geq 5) = 0.39$.

8.5.5 A “100-year water,” or flood, is the water level that is exceeded once in a 100 years (on average). Suppose that the threatening water levels occur once a year and have a normal distribution. Suppose also that at some location the 100-year water means the level of 30 feet above average. What is the 10,000-year water level?

8.5.6 Find γ_3 —the skewness of a lognormal distribution with parameters μ and σ^2 .

- 8.5.7** Assume that X_1 and X_2 are independent, with $N(3, 6)$ and $N(-1, 2)$ distributions, respectively. Find: **(i)** $P(3X_1 - 2X_2 \geq 14)$. **(ii)** $P(X_1 < X_2)$.
- 8.5.8** Assume that variables X_1, X_2, X_3 are independent and each has $N(2, 1)$ distribution. Moreover let Y_1, Y_2, Y_3 , also be independent and each have $N(2, 3)$ distribution. Find $P(3X_1 + 2X_2 + X_3 > Y_1 + 2Y_2 + Y_3)$.
- 8.5.9** Assume that variables X_1 and X_2 have a bivariate normal distribution with $E(X_1) = 3, E(X_2) = 2, \text{Var}(X_1) = 4, \text{Var}(X_2) = 1$ and $\rho = -0.6$. Find: **(i)** $P\{X_1 \leq 4 | X_2 = 3\}$. **(ii)** $P\{|X_2 - 1| \geq 1.5 | X_1 = 2\}$.
- 8.5.10** Let variables X and Y be the weight and height of newborn children in the United States. The joint distribution of X and Y is bivariate normal with $\mu_X = 20, \sigma_X = 1.5$ inches, $\mu_Y = 7.7, \sigma_Y = 2$ pounds, and $\rho = 0.8$. Find: **(i)** The expected weight of a newborn child with a height 21 inches. **(ii)** The expected height of a child whose weight is 6 pounds.

8.6 BETA DISTRIBUTION

The family of *beta distributions* defined below is known for its usefulness in modeling researchers' uncertainty about the unknown probability p of some event.

Definition 8.6.1 A random variable X has *beta distribution* with parameters $\alpha > 0$ and $\beta > 0$, $X \sim \text{BETA}(\alpha, \beta)$, if the density of X equals

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} \quad (8.68)$$

for $0 \leq x \leq 1$ and $f(x) = 0$ otherwise. Here Γ is the function defined by (8.46). $\square$

First, we will check that (8.68) is indeed a density:

$$\begin{aligned} \Gamma(\alpha)\Gamma(\beta) &= \left(\int_0^\infty u^{\alpha-1} e^{-u} du \right) \left(\int_0^\infty v^{\beta-1} e^{-v} dv \right) \\ &= \int_0^\infty \int_0^\infty u^{\alpha-1} v^{\beta-1} e^{-(u+v)} du dv. \end{aligned} \quad (8.69)$$

Introducing

$$z = u + v \quad \text{and} \quad x = \frac{u}{u + v},$$

we have $0 \leq x \leq 1, 0 \leq z \leq \infty$. Since $u = xz$ and $v = z(1-x)$, the Jacobian equals z . Consequently, after substitution to (8.69), the variables separate, and we obtain

$$\begin{aligned} \Gamma(\alpha)\Gamma(\beta) &= \int_0^\infty z^{\alpha-1} z^{\beta-1} e^{-z} z dz \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx \\ &= \Gamma(\alpha + \beta) \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx, \end{aligned}$$

as was to be shown.

Once we know that

$$\int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)},$$

we can easily compute the moments of the beta distribution. If $X \sim \text{BETA}(\alpha, \beta)$, then

$$\begin{aligned} E(X^k) &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^{k+\alpha-1}(1-x)^{\beta-1} dx \\ &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \times \frac{\Gamma(k + \alpha)\Gamma(\beta)}{\Gamma(k + \alpha + \beta)} = \frac{\Gamma(\alpha + \beta)\Gamma(k + \alpha)}{\Gamma(\alpha)\Gamma(k + \alpha + \beta)}. \end{aligned}$$

In particular, using formula (8.47), for $k = 1$ and $k = 2$, we obtain

$$E(X) = \frac{\alpha}{\alpha + \beta} \quad \text{and} \quad E(X^2) = \frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)},$$

so that

$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}. \quad (8.70)$$

The various shapes of the beta distribution are illustrated in Figure 8.7. If $\alpha > 1, \beta > 1$, the distribution is bell-shaped, with the peak becoming more narrow for larger α and/or β . For $\alpha = \beta = 1$, the distribution is uniform. If $\alpha < 1$ and $\beta < 1$, the distribution is U-shaped, whereas if $\alpha < 1, \beta \geq 1$ or $\alpha \geq 1, \beta < 1$, the distribution is J-shaped.

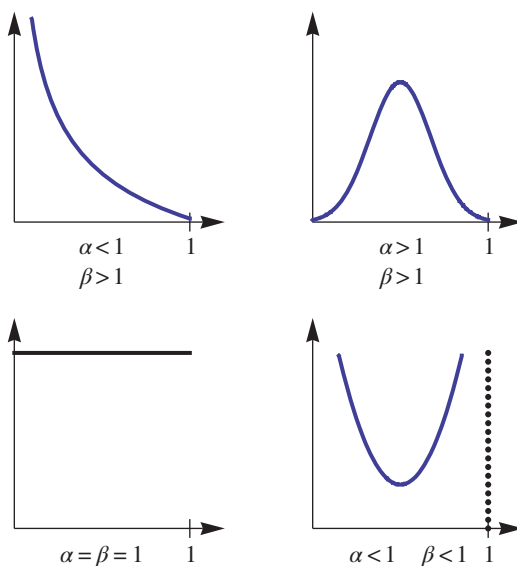


Figure 8.7 Shapes of beta distributions.

As mentioned, the most typical application of a beta distribution is when we consider a binomial distribution with an unknown p , and represent knowledge (or uncertainty) about p by assuming that p is random with a beta distribution.

■ EXAMPLE 8.20

In some Bernoulli trials p is random with distribution $\text{BETA}(\alpha, \beta)$. We know that there were so far 3 successes in 4 trials. What is the probability of 2 successes in the next 3 trials?

The combined data and event in question is “3 successes in 4 trials and 2 successes in the next 3 trials” (which is *not* the same as 5 successes in 7 trials. Why?). Conditioning on p , we obtain the probability

$$\binom{4}{3} p^3(1-p) \binom{3}{2} p^2(1-p) = 12p^5(1-p)^2.$$

Taking expectation with respect to $p \sim \text{BETA}(\alpha, \beta)$, we have, for the expected probability $E(2, 3)$ of 2 successes in 3 trials

$$\begin{aligned} E(2, 3) &= \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 12p^5(1-p)^2 p^{\alpha-1}(1-p)^{\beta-1} dp \\ &= 12 \times \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \times \frac{\Gamma(5 + \alpha)\Gamma(2 + \beta)}{\Gamma(7 + \alpha + \beta)}. \end{aligned}$$

Using formula (8.47), which asserts that $\Gamma(t) = (t-1)\Gamma(t-1)$, we obtain, after some algebra,

$$E(2, 3) = 12 \times \frac{\alpha(\alpha+1) \cdots (\alpha+4)\beta(\beta+1)}{(\alpha+\beta)(\alpha+\beta+1) \cdots (\alpha+\beta+6)}.$$

■ EXAMPLE 8.21

To make the situation still more realistic, consider a client who comes to a statistical consulting center for help. It turns out that she is currently negotiating the purchase of a large shipment of some merchandise. Some of the items conform to the specification, and some do not (let us call them defective). Suppose that testing the quality of all items in the lot is not possible because of the prohibitive cost of such an operation, or because the process of testing is destructive. Some tests have been made, though, and the data are such that among 25 tested items 3 were found to be defective. What can one reasonably say about the fraction of defective items in the whole lot?

SOLUTION. Suppose that the lot in question is not the first lot purchased by the client. It turns out that she has been buying the same kinds of merchandise from the same company (or from other companies). In her experience, as judged by tests, data on returns, customer complaints, and so on, she estimates that on average, the percentages of items below specification (defectives) in various lots purchased were about 12%, with variability of about 2% in either direction.

The problem of the client at the statistical consulting center can be formulated as follows: First, we determine the chances that the fraction p of defective items in the lot whose purchase is presently being negotiated is below 10%. Then we evaluate $P\{p < 0.1\}$ in light of the tests of the lot under consideration (3 out of 25 below specification), using the experience accumulated in the past. If we accept the assumption that the fractions in various lots follow a beta distribution, we have for the mean

$$\frac{\alpha}{\alpha + \beta} = 0.12. \quad (8.71)$$

Next, the relatively small variability ($\pm 2\%$) suggests that we have $\alpha > 1$ and $\beta > 1$. Consequently, the density of beta distribution is bell-shaped. Using the three-sigma rule, we take $3\sigma \approx 0.02$, and using (8.70), we obtain another equation for α and β :

$$\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \left(\frac{0.02}{3}\right)^2. \quad (8.72)$$

The solution of (8.71) and (8.72) is $\alpha^* = 285$, $\beta^* = 2,090$.

We have assumed here that the fraction p of defective items is a random variable whose distribution is $\text{BETA}(\alpha^*, \beta^*)$. If no testing of the currently negotiated lot was made, the answer would be

$$P\{p < 0.1\} = \frac{\Gamma(\alpha^* + \beta^*)}{\Gamma(\alpha^*)\Gamma(\beta^*)} \int_0^{0.1} x^{\alpha^*-1} (1-x)^{\beta^*-1} dx.$$

However, we have additional information, namely that event $A =$ “three defective in a sample of 25” occurred. Consequently, we should integrate the conditional density of p given A . We have

$$P(A|p) = \binom{25}{3} p^3 (1-p)^{22}.$$

The conditional density is obtained from Bayes’ formula, with the summation replaced by integration. Schematically

$$\varphi_p(x|A) = \frac{P(A|x)\varphi_p(x)}{\int P(A|u)\varphi_p(u) du}, \quad (8.73)$$

where $\varphi_p(x)$ and $\varphi_p(x|A)$ are the unconditional and conditional densities of p at x .

Typical reasoning is that the denominator in (8.73) is independent of x , and both sides represent densities in x . Collecting the terms that depend on x , and representing all other factors as a constant, the right-hand side of (8.73) is of the form

$$Cx^3(1-x)^{22}x^{\alpha^*-1}(1-x)^{\beta^*-1}, \quad (8.74)$$

so it is a beta density with parameters $\alpha^* + 3$ and $\beta^* + 22$. This gives

$$C = \frac{\Gamma(\alpha^* + \beta^* + 25)}{\Gamma(\alpha^* + 3)\Gamma(\beta^* + 22)},$$

and we get the answer to the original question by integrating (8.74) between 0 and 0.1.

Observe that the analysis above has some interesting consequences. We know that if the distribution of p is $\text{BETA}(\alpha, \beta)$ and we observe k successes in n trials (event A), then the conditional distribution of p given A is $\text{BETA}(\alpha + k, \beta + n - k)$ —the family of beta distributions that constitutes the conjugate priors for binomial data. We will return to those problems in Chapter 16.

PROBLEMS

- 8.6.1** Let X have a distribution $\text{BETA}(\alpha, \beta)$. Find: **(i)** the distribution of $Y = 1 - X$. **(ii)** $E\{X^k(1-X)^m\}$.
- 8.6.2** Students are late to a particular class with probability that has $\text{BETA}(2, 20)$ distribution. Find the probability that in a class of 25 students no more than 2 students will be late on a given day.
- 8.6.3** Let X have a symmetric beta distribution. Find α and β if the coefficient of variation (ratio of standard deviation and the mean) is k . Does a solution exist for any k ?
- 8.6.4** Let $X_1, X_2, \dots$ be iid with an exponential distribution. For any positive m and n find the distribution of the ratio

$$T_{m,n} = \frac{X_1 + \dots + X_m}{X_1 + \dots + X_{m+n}}.$$